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List of Abbreviations

BMW – Bayerische Motoren Werke

CPC – Cooperative Patent Classification

DDD – Difference-in-Differences-in-Differences (Triple Difference-in-Differences)

DOCDB – Documentation Database (EPO worldwide patent family database)

DTC – Directed Technical Change

EPA – Environmental Protection Agency

EPO – European Patent Office

EU – European Union

EV – Electric Vehicle

FE – Fixed Effects

GM – General Motors

HEV – Hybrid Electric Vehicle

ICE – Internal Combustion Engine

IPC – International Patent Classification

NPL – Non-Patent Literature

OLS – Ordinary Least Squares

PATSTAT – Patent Statistical Database (EPO Worldwide Patent Statistical Database)

PCT – Patent Cooperation Treaty

PPML – Poisson Pseudo-Maximum Likelihood

UK – United Kingdom

US / USA – United States / United States of America

USD – United States Dollar

USPC – United States Patent Classification

USPTO – United States Patent and Trademark Office

VW – Volkswagen

WIPO – World Intellectual Property Organization

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1. Introduction

The Volkswagen emissions scandal (Dieselgate), which broke in 2015, is one of the largest corporate scandals in recent history, stemming from violations of environmental standards. Volkswagen (VW) was required to pay substantial fines and settlements after it was revealed that the company had systematically concealed that many of its vehicle models failed to comply with environmental regulations, despite passing official emissions tests. This scandal can be viewed not only from a legal perspective but also from an innovation standpoint: rather than achieving genuine technological progress, the company concealed its shortcomings to avoid losing sales. The financial consequences of the fraudulent strategy amounted to approximately USD 35 billion in fines and settlements, which should, in principle, have discouraged Volkswagen from pursuing such a strategy in the future. The company's annual reports (Volkswagen Group, 2015, 2020a) document a clear shift toward the electrification of its vehicle fleet: while the 2015 annual report recorded no fully electric vehicle (EV) deliveries worldwide, by 2020 the company had delivered 231,624 fully electric vehicles globally. This suggests that, in the aftermath of the scandal, Volkswagen implemented reforms in its innovation strategy by reallocating part of its R&D efforts towards environmentally friendly automotive technologies. In this thesis, I investigate how the Dieselgate scandal affected innovation output among German automobile manufacturers—Volkswagen, BMW, and Daimler. These firms are treated as a single group for reasons established in the existing literature, as discussed in this and later sections. More specifically, I examine whether the reputational and financial shocks that directly affected unit sales and consumer valuations of German automakers' cars in the United States (Bachmann et al., 2023) changed the innovation path of German automakers: from innovations associated with internal combustion engine (ICE, dirty technologies) cars to those driven by electric propulsion technologies (green). Next, I elaborate on the scandal's context.

Dieselgate is a major scandal associated mainly with VW that broke publicly in the United States when the U.S. Environmental Protection Agency issued a formal Notice of Violation against Volkswagen on September 18, 2015 (U.S. EPA, 2026). The EPA discovered that Volkswagen had installed illegal "defeat device" software in diesel cars. This device could recognize when the car was being emissions-tested and when it was actually being driven under normal conditions. In the first case, the device was deliberately activated to reduce the concentration of certain exhaust gases and chemicals so as to meet EPA emissions standards. On the road, the cars emitted nitrogen oxides up to 40 times above the legal limit (U.S. EPA,

2026). VW shares lost up to 37% of their value in the days immediately following the September 18, 2015, revelation (Reuters, 2018), and VW CEO Martin Winterkorn resigned shortly after the scandal broke, on September 23, 2015. By 2021, VW confirmed the diesel scandal had cost a total of $34.7 billion in fines and settlements. More than half was paid to US authorities: a $14.7 billion civil agreement with EPA, California, and other U.S. federal agencies in July 2016 (NBC News, 2017), and a $2.8 billion criminal fine paid to the U.S. Department of Justice — the largest criminal fine ever imposed on a carmaker in the U.S. (NBC News, 2017). A $1.18 billion fine was imposed on Volkswagen by German authorities in June 2018 (Deutsche Welle, 2018). In addition, VW paid an €830 million settlement in Germany to the Federation of German Consumer Organisations (Volkswagen Group, 2015, 2020b). Depending on model and age, VW compensated roughly 260,000 German customers with between €1,350 and €6,257 each. Other countries, including Italy, the United Kingdom, South Korea, Canada, France, and Australia, also launched investigations (BBC, 2015; Reuters, 2024). In total, 11 million vehicles worldwide were equipped with the defeat device software (NBC News, 2017). Notably, VW recalled 8.5 million vehicles in Europe, amounting to 77% of all affected cars. The recall was substantive rather than a mere formality, and European authorities continued to monitor the situation closely. For instance, on October 11, 2017, the European Parliament discussed the fact that approximately 3 million VW Group vehicles equipped with defeat devices were still on European roads (European Parliament, 2017). The scale of the fines and settlements, and their geographical spread, suggests that any resulting changes in VW's innovation behaviour are likely to be global rather than confined to the US market. For this reason, my research is not restricted to US patents alone but considers innovations developed globally.

A further notable aspect of Dieselgate is that it also significantly affected other German automakers in the United States (Bachmann et al., 2023). The demand model in Bachmann et al. (2023) reveals that consumer valuations fell by an average of $3,057 for VW Group vehicles and $2,057 for non-VW German manufacturers' vehicles in the first year following the scandal (through August 2016). The spillover effect alone would have reduced non-VW German automakers' annual U.S. unit sales by 244,607 vehicles (-34.6%), while the substitution effect generated an increase of 77,342 units (+11%) in non-VW German automakers' sales. After netting out the substitution effect, the net loss in unit sales amounts to -23.5%, worth $7.74 billion. The authors of the paper focused exclusively on the U.S. market, meaning the losses of the non-VW German automakers could be even larger if other markets were taken into account. In total, the effect for the German auto industry could amount to more than $40 billion. The

authors also show that the scandal reduced non-VW German automakers' 12-month sales growth relative to non-German makers operating in the U.S. through a so-called country-of-origin reputation channel. The hypothesis of pure substitution in favor of non-diesel cars (German automakers were major producers of diesel cars) also failed, since the sales of non-diesel vehicles decreased by a similar number of percentage points relative to non-German makers. The importance of the results in Bachmann et al. (2023) lies in two implications. First, it gives me a basis to hypothesize about the likely differences in innovation policy at the level where brands are grouped by country of origin. Second, since the non-VW German automakers bear enormous involuntary reputation costs due to the scandal initially associated with the VW Group, a strong economic rationale arises for those firms to invest in visible, credible signals of their commitment to environmental protection and green technologies. The VW Group was also strongly affected, which I interpret as a valid reason to treat VW Group and the non-VW German makers as a single group. The paper also notes that German automakers actively cultivated "German engineering" as a shared identity in their U.S. advertising, and the authors cite VW's own slogan "Das Auto". This identity of superior German engineering was directly attacked by the scandal. I argue that advertising alone would not be sufficient for German makers to restore their reputation, and that substantiating this claim would also require genuine technological advancement.

Barth et al. (2022) find support for the reputational spillover effect caused by Dieselgate in capital markets: Daimler and BMW suffered the largest individual abnormal equity losses among all competitors worldwide, while European competitors as a group showed significantly negative spillover effects for up to ten days afterwards. In contrast, the effects were insignificant for non-European firms. These findings reflect an "industry generalization" pattern, whereby investors do not distinguish between individual firms but instead react to a trait common to the group — in this case, the identity expressed by the "Das Auto" slogan.

The core of my empirical approach is the triple difference-in-differences (DDD) model, which captures the scandal's effect on the yearly outcome gap between dirty and green technologies for German makers after the scandal. I treat non-German makers as a control group and use patent data from the PATSTAT (Patent Statistical Database) database to measure the yearly innovation outcome over the period 2010 - 2019. PPML (Poisson Pseudo Maximum Likelihood) is used to model the conditional mean of the innovation outcome data nonlinearly, and a linearized model is proposed as a robustness check. Since the scandal's effect can differ across years, I estimate dynamic DDD coefficients. This analysis reveals fading effects and unevenness in the economic impact of Dieselgate. A disadvantage of the dynamic approach lies

in the difficulty of interpreting the scandal's influence when a significant impact is not detected in every year. The static analysis makes it easier to assess the scandal's impact over the entire post-scandal timeframe.

The results of this thesis make it possible to assess how the discovery of environmental regulation violations, accompanied by substantial reputational consequences, affects firms' innovation strategies. In the context of the automotive industry, the thesis provides policymakers with insights into the extent to which public exposure of automakers' non-compliance with environmental standards encourages them to shift towards alternative technologies. Furthermore, by identifying broader patterns in technological development, the paper investigates the presence of internal spillover effects of Dieselgate—changes that arise from strategic decisions made within firms themselves, rather than external consequences such as regulatory fines or declining vehicle sales, which reflect market and societal reactions. The study finds that the gap in patent counts between the two technology branches was reduced, on average, by 48.8% after the scandal in favor of green technologies, meaning the growing share of green patents in the entire patent pool. I also find that the gap in the accumulated number of jurisdictions where a technology was patented (family size) was reduced in favor of green patents by around 50%, meaning green innovations are being patented on an increasingly global scale alongside the growing absolute number of patents.

The remainder of the thesis is structured as follows: I proceed with a literature review, discussing studies that lay the theoretical background for my thesis and present approaches to measuring innovation through patents. In the third section, I focus on the data sources and how the panel dataset was constructed. Section four is devoted to the methodology I employ to identify the scandal effect, and in Section five, the results of the research are discussed, including possible explanations and alternative interpretations. The thesis concludes in Section 6.

2. Literature review

In this thesis, I examine how the relationship between the two classes of innovation by German makers developed after the scandal broke out. The Directed Technical Change (DTC) theory provides a solid background for understanding how producers of different types of technologies decide which types of technologies to produce. Under the classical DTC by

Acemoglu (2002), the focus is on the relative profitability of creating technologies (machines) of two different types. The core channels influencing this relative profitability are the price effect and the market size effect. The market size effect arises from an increase in relative factor supply: under the classic DTC, an increase in the number of workers (or capital, skilled labor, or land) associated with a given type of innovation (machines) stimulates demand for those machines. Hence, the profitability of producing that type of machine will rise relative to the second type of machine. The relative prices in this case will fall (price effect), thereby pushing the relative profitability downwards. Which of the two effects is stronger depends on the elasticity of substitution between the two types of production factors for which these two types of machines are produced. If the factors are relatively easily substitutable (elasticity of substitution greater than one), then the market size effect increasingly dominates the price effect, stimulating the production of the abundant factor. In the reverse case, when the factors are gross complements (elasticity close to zero), the price effect increasingly dominates, fostering investment in the scarce factor. Hémous & Olsen (2021), synthesizing Acemoglu et al. (2012), extend the same DTC mechanism to a setting in which clean and dirty energy are treated as the final-product production factors and introduce a third channel: the direct productivity effect, captured by the ratio of accumulated technology stocks. Under the assumption that clean and dirty inputs are gross substitutes, this term is self-reinforcing: the sector with the larger knowledge stock is more profitable and attracts more researchers (higher R&D). The result is directional path dependence under laissez-faire (no market interventions): an initial advantage in the dirty sector locks in innovation in that sector, and the government would have to implement either research subsidies for clean innovation — preserving earnings for producers of clean technologies while making the final good more affordable — or a carbon tax that lowers the profitability of dirty innovation.

A complementary strand of the DTC literature uses calibrated structural models rather than estimations based on real data to study innovation redirection. Gerlagh (2008) develops an endogenous growth model in which a carbon tax permanently raises the relative cost of dirty energy production and generates a sustained reallocation of R&D toward carbon-energy-saving technology. This paper provides a useful theoretical counterpoint for the present analysis: it illustrates the magnitude and persistence of redirection predicted under an ongoing price shock, against which a reputational shock without a comparable price mechanism can be contrasted.

I interpret Dieselgate as an exogenous shock to the automotive industry, which I view here as a producer of both green (clean) technologies, aimed at electric vehicles, and dirty technologies, aimed at cars powered by internal combustion engines. The reputational collapse

and demand reaction documented by Bachmann et al. (2023) could foster a contraction in the market size for ICE technologies (market size effect), thereby shifting innovation incentives toward clean technology. Che et al. (2020) provide evidence of a reduced willingness to pay for Volkswagen cars. They use the EPA's announcement as an exogenous shock to eBay used-car prices and show that Volkswagen diesel models not implicated in the violation (model years 2000–2008) suffered a 14% price drop — a larger drop than that experienced by the models actually implicated in the violation (10.8%) — despite carrying none of the latter's compliance risk or recall costs. German automakers are therefore expected to have redirected R&D from ICE toward EV technologies following 2015.

An important methodological reference for patent selection is Aghion et al. (2016). The researchers pose the question: "Can directed technical change be used to combat climate change? Specifically, do higher (tax-inclusive) fuel prices redirect innovation in the auto industry away from dirty (ICE) and toward clean (electric, hybrid, hydrogen) technologies, and does path dependence in innovation histories strengthen or counteract this effect?" They find that "clean innovation is stimulated by increases in fuel prices, whereas dirty innovation is depressed". They also find evidence for the so-called "path dependency": firms that are historically prone to clean innovation are more likely to direct their research efforts toward clean innovation in the future. The authors use PATSTAT data on patents and classify them into three groups: clean, dirty, and grey patents. Although they use IPC (International Patent Classification) patent classes, the CPC (Cooperative Patent Classification) classes I use for classification are an extension of IPC (European Patent Office, 2025).

In my framework, I measure automakers' innovative productivity using patent counts — which do not account for the private or social value of each patented innovation — as well as indicators of the patents' qualitative characteristics. The key advantages of this approach are that patents are available for all firms and individuals, contain details on the technological area, the inventor, and the assignee, and are by definition related to inventiveness (Griliches, 1998). The existence of a patent can be considered evidence of innovation activity of greater or lesser magnitude (value), and this nature of patents is non-speculative. I also cover PATSTAT-related advantages of patent counts in the third section of this thesis.

Despite the advantages of patent counts, they measure the 'size' of innovations that have been or are being developed, but neglect the qualitative aspect of innovations. Patent counts treat every patent as equivalent, but patent value is heterogeneous. Harhoff, Scherer & Vopel (2003) find that the private commercial value of patents varies substantially among patents owned by German firms. They measure the value of a patent right as 'the price for which the

original inventor would be willing to sell the patent right'. Lanjouw, Pakes & Putnam (1998) argue that a large variance in patent values causes a substantial degree of noise in patent count measures, making identification in innovation studies with patent counts less reliable. The shortfalls of patent data were also demonstrated using time-series data from Schankerman & Pakes (1986), who found that in the UK, France, and Germany, over 1965-1975, count indices and value indices moved in opposite directions. In light of this value heterogeneity and the documented tension between patent counts and quality, I extend the analysis to examine the qualitative aspects of patents owned by automotive firms.

One of the widely accepted approaches to measuring the qualitative dimension of innovations is to consider two of each patent's features: patent renewals and family size. According to Lanjouw, Pakes & Putnam (1998), the logic of patent renewal rests on the fact that each patent right must be periodically prolonged and that patentees must pay periodic renewal fees; otherwise, they lose the patent right and forfeit the competitive advantage that patent protection provides. The second aspect concerns the fact that patentees can patent the same invention in many countries (jurisdictions), securing their competitive advantage not only over time but also geographically. Therefore, the same or very similar invention can be protected by patents in a few or even dozens of independent patent authorities. This fact gave rise to the terms 'family of patents' and 'family size' in the patent literature discussed in this work. Renewals and family size are assumed to convey important information about patent quality, because each additional country filing carries translation, attorney, and renewal costs — the more jurisdictions covered, the higher the total renewal costs. These non-negligible costs allow us to think of patenting as a deliberate decision-making process undertaken by rational actors (Putnam, 1996). In line with Putnam (1996), an invention will only be patented in a jurisdiction if net returns are positive. The number of countries a patent is filed in, therefore, reveals the inventor's expectation of its commercial value. The logic behind using renewals and 'families' in qualitative analysis is concisely summarized in Lanjouw, Pakes & Putnam (1998) in the following statement: "the private value of a patent is determined by the difference in the returns that would accrue to the innovation with and without patent protection" (p.2).

Although both renewals and families can be used to proxy for patent value, for my research design, I choose family size alone as the patent value indicator. For this decision, I offer three reasons. The first reason is timing: family information is available within a few years of first filing. Under the Patent Cooperation Treaty (PCT), inventors have at most 30 months from the priority date (filing date in the priority jurisdiction) to file patent applications in additional jurisdictions worldwide (WIPO, 2002). Specifically, PCT Article 22(1), amended

with effect from 1 April 2002, stipulates that national phase requirements must be met no later than 30 months from the priority date. This rule remained unchanged throughout the 2010 – 2020 period covered by this analysis. Renewal trajectories require 10–20 years to be informative. For example, Lanjouw, Pakes & Putnam (1998) use patents with a duration of up to 20 years before lapse to estimate value dynamics. Dieselgate occurred in September 2015; the post-period is 2015–2020. Renewal data cannot speak to whether post-2015 patents are more valuable, because those patents have not aged enough. The third reason is that the DOCDB (Documentation Database) family data within PATSTAT cover the full sample uniformly. Renewal data vary in completeness and format across jurisdictions. Moreover, renewals of the same patent in one jurisdiction alongside the termination of patent rights without a subsequent prolongation in another can further complicate the interpretation of the renewal event.

Next, I present examples of papers that empirically confirm the strong correlation between family size and patent value. Harhoff, Scherer & Vopel (2003) surveyed German patent-holders, asking them to assess the asset value of their patent rights and eliciting value estimates via a single counterfactual question about minimum compensation. Based on an ordered probit regression, they found that a doubling of family size corresponds to ~75% higher expected value (author's calculation). The specification controls for forward and backward citations, references to patent- and non-patent literature, opposition outcomes, and technology-field fixed effects, mitigating concerns about omitted variable bias across sectors. However, a causal interpretation requires caution: as the authors acknowledge, family size is itself a choice variable, with patent holders filing across more jurisdictions precisely for inventions they expect to be highly valuable. The causal arrow therefore runs from anticipated value to family size, rather than the reverse. It reflects the patent holder's own private assessment of commercial potential at the time of filing, making family size an economically grounded signal of patent value. In another example (Gambardella, Harhoff & Verspagen, 2006), family size is called one of the 'traditional indicators' of patent value, along with forward citations, backward citations, the number of claims in the patent, and whether the patent was opposed by other parties. The authors of this article use the PatVal-EU survey and other datasets to explore the impact of non-traditional determinants, such as inventor characteristics, on the value of patents. Patent-value data, collected in the form of intervals (e.g., less than €30K; €30-100K), were collected from inventors' answers. The results of the paper support the statement that patents filed in more jurisdictions are associated with being placed in higher value intervals. The authors also mention that 'our measure reproduces most closely the patent family size'. This statement is

supported by the estimation of the ordered probit model, where family size has the largest per unit (unit is a value interval) effect among count variables (forward citations, backward citations, claims) and is significant at the 1% level. Since the geography of inventors in Gambardella, Harhoff & Verspagen (2006) includes France, Germany, Italy, the Netherlands, Spain, and the UK, I can consider family size a valid indicator of a patent's value on a European scale, and not only for a certain type of inventor or country. In this regard, I turn to research using data from the USA. Lanjouw & Schankerman (2004) developed a composite index of patent quality using the number of claims, forward citations, backward citations, and family size, as part of an examination of the relationships between research productivity, market size, and technological exhaustion at the firm level. The authors use data on 100,000 patents filed by manufacturing firms from seven industries (Drugs, Biotech, Other Health, Chemicals, Computers, Electronics, and Mechanical) in the US during the period 1975 – 1993. The authors find that family size is a statistically significant indicator of patent quality (at the 5% level) for five industries, including Mechanical. The authors acknowledge "this aggregation updates an earlier aggregation by Adam Jaffe". This aggregation is specified in the supplemental materials to Jaffe, Fogarty and Banks (1998). These supplemental materials explicitly list USPC (US Patent Classification) classes 123 (Internal-Combustion Engines), 180 (Motor Vehicles), 280 (Land Vehicles), 296 (Land Vehicles: Bodies and Tops), 298 (Land Vehicles: Dumping), 301 (Land Vehicles: Wheels and Axles), 310 (Electrical Generator or Motor Structure) and 388 (Electricity: Motor Control Systems) among the Mechanical classes. The listed classes are directly associated with the automotive industry considered in this thesis. Moreover, the authors of Lanjouw & Schankerman (2004) suggest that family size is the closest proxy for the monetary value of the patent, because the increase in the number of patents issued for the same invention implies an increase in patenting fees, which is only a rational decision if expected gains also rise. Building on these findings regarding the significance of family size for patent quality in the USA, together with the broader evidence on the relationship between family size and patent value, I treat family size as an adequate proxy for patent value.

The forward and backward citations mentioned earlier are additional patent attributes that signal an invention's quality. A forward citation occurs when a given patent is cited by subsequent patents (Trajtenberg, 1990). Some researchers have used the count of forward citations to overcome the disadvantages of a pure patent-count approach. Trajtenberg (1990) provides three reasons for using forward citations. The first is that there is a legal incentive to cite all and only relevant patents, since citations limit the scope of property rights, and the examiner actively enforces this. This implies that a patent is unlikely to be cited without a valid

reason. The second reason builds on the assertion that a citing patent attests that the cited one opened a technologically successful line of innovation. R&D costs preceding subsequent patenting by profit-seeking firms signal the economic success of the cited patent. Trajtenberg (1990) finds that in computed tomography, simple patent counts correlate strongly with R&D inputs, with a minimum correlation of 0.831 for a 1-year lag (Table 6), but are essentially uncorrelated with innovation value (correlations with the independent social measures of innovation value range from -0.283 to 0.131 across all tested lags), none statistically significant (Table 3). Citation-weighted counts change this: correlations with social value indicators rise to 0.755 – 0.813 depending on lag (Tables 3 and 5). This paper offers a canonical example of how forward citations can be combined with historical patent counts to proxy for the value of innovation. This finding is especially relevant for this thesis because it shows that forward citations can approximate patent value, much like family size. However, this paper does not offer a clear guideline on the minimum citation accumulation time, and the answer to this question largely depends on the industry under consideration. Lanjouw & Schankerman (2004) claim "forward citations are related most directly to technological importance". They find that forward citations (5-year span) are a statistically significant indicator of patent quality and have a larger weight in the Patent Quality Index (14.7) than family size (8.3). This paper is especially relevant here because it provides an example of the time period after the patent application date within which forward citations occur. The authors use two variables with a 5- and 10-year citation span. Since the paper is based on a large dataset of patents belonging to US manufacturers, this provides a useful starting point for determining a proper time span for the automotive patents used in my analysis. Backward citations are prior patents cited within an application. In the regression of patent value on observable characteristics in Harhoff, Scherer and Vopel (2003), backward citations enter as a positive and significant predictor alongside family size and forward citations, providing direct empirical validation that backward citations signal higher private patent value.

References to non-patent literature (NPL) are citations to scientific and technical publications outside the patent system, such as journal articles. They indicate an invention's science linkage, i.e., how closely it draws on upstream research rather than purely applied prior patents (Trajtenberg, 1990; Harhoff, Scherer & Vopel, 2003). Harhoff, Scherer & Vopel (2003) treat NPL references as one of several established value proxies, including them alongside forward and backward citations. As a quality indicator, NPL counts are directly observable and require no subjective valuation. In the next section, I discuss how the patent data were derived from PATSTAT and transformed into a panel suitable for my analysis.

3. Data

I use the PATSTAT Global database as a source of patent-application data and patent-application attributes (family size, backward citations, forward citations, non-patent literature), and information on firms that serve as patent applicants. It is important to clarify the difference between patent applications and patents. The former is a document filed in a patent office and subsequently examined by the relevant authorities. As a result of this examination, the innovation is either granted a patent or rejected. I chose patent applications as the unit of observation, since applications signal a maker's intention to innovate in a given field. For simplicity, I use the word "patents" as shorthand for "patent applications".

One property of patents that shapes the advantage of patent counts as an innovation measure was not mentioned in the literature review. Additional data in the PATSTAT database allow for the explicit differentiation of patents by their innovation importance. The principle of continuations represents a "priority-like" relationship between an earlier (parent) application and a subsequent (child) application filed with the same domestic patent authority. This allows researchers to refine the patent dataset by excluding continuation patents and retaining only core, independent patent counts. For testing DTC predictions (technological path dependence) specifically, patent counts measure exactly what the theory addresses: where inventive effort is concentrated across the technology field, capturing the extensive innovation margin driven by Dieselgate. In other words, counts answer 'how much' and 'where'.

In total, I collected 31,557 patents filed in the US by automakers operating in the American market between 2010 and 2019. Table A.1 from Bachmann et al. (2023) served as a reference for the brands presented in the American market. I used these brand names to identify as many automakers as possible, filtering for corporate entities and excluding individuals with matching surnames. Next, I filtered the names according to their relevance to the American market and their significance. For example, names like “FORD-WERKE” or “AUDI NSU AUTO UNION” were excluded since the former operated exclusively in Germany, and the latter no longer exists. Next, I eliminated those entities that had fewer than 400 patents. This threshold was chosen for several reasons. First, entities at the top of the list accumulated tens of thousands or even hundreds of thousands of patents and were associated with renowned automaker brand names (Toyota, General Motors, Ford, BMW; see the full list in Table 1), whereas the enterprises with very few patents did not represent major automakers or their

prominent subsidiaries. For this reason, I did not consider these businesses direct competitors to the incumbents in the car market. The second reason was to facilitate the ownership-structure analysis. In this thesis, innovation policy decisions are assumed to depend on ownership structure. For instance, Audi belongs to Volkswagen. Although they are two different brands, I consider them a single brand group united under the name "Volkswagen Group". For a large number of small entities, information about changes in ownership structure is often unpublished, which makes brand-group assignment impossible. For the remaining firm names, I conducted a manual exclusion. Finally, I allocated the firms to 18 brand groups. Table 1 presents the full list of brand groups and the PATSTAT psn_names associated with each group. For each group, the number of patents per year and per innovation type (green or dirty) was defined as the sum of the number of corresponding patents for each included company. psn_name is an identifier of a person's name in PATSTAT (the PATSTAT Standardised Name). A person can have different types of relations to a patent (e.g., applicant, inventor). In this table, only applicants are considered.

Table 1. Brand Groups, psn_name mapping.

Brand group	**Consolidated psn_name(s) (applicants)**	**Country**	**Patent Counts**
Volkswagen Group	VOLKSWAGEN, AUDI, PORSCHE	Germany	1,507
Daimler	DAIMLER	Germany	281
BMW	BMW (BAYERISCHE MOTOREN WERKE)	Germany	713
Ford	FORD GLOBAL TECHNOLOGIES, FORD MOTOR COMPANY, FORD-WERKE, LINCOLN GLOBAL, LINCOLN	United States	5,865
General Motors	GM (GENERAL MOTORS CORPORATION), GM GLOBAL TECHNOLOGY OPERATIONS, GENERAL MOTORS RESEARCH CORPORATION, SAIC GENERAL MOTORS CORPORATION, SHANGHAI GENERAL MOTORS COMPANY, SAIC-GM-WULING AUTOMOBILE COMPANY	United States	3,929

Brand group	Consolidated psn_name(s) (applicants)	Country	Patent Counts
Toyota	TOYOTA MOTOR CORPORATION, TOYOTA INDUSTRIES CORPORATION, TOYOTA MOTOR ENGINEERING & MANUFACTURING NORTH AMERICA, TOYOTA BOSHOKU CORPORATION, TOYOTA CHUO KENKYUSHO, TOYOTA SHATAI, TOYOTA MOTOR EUROPE, TOYOTA INFOTECHNOLOGY CENTER COMPANY, TOYOTA CENTRAL RES & DEV LAB, TOYOTA MOTOR EAST JAPAN	Japan	7,747
Honda	HONDA MOTOR COMPANY, HONDA ACCESS CORPORATION, HONDA LOCK MANUFACTURING COMPANY, HONDA LOCK	Japan	2,817
Nissan	NISSAN MOTOR COMPANY, NISSAN NORTH AMERICA	Japan	1,208
Mitsubishi	MITSUBISHI MOTORS CORPORATION, MITSUBISHI HEAVY INDUSTRIES ENGINE & TURBOCHARGER, MITSUBISHI HEAVY INDUSTRIES THERMAL SYSTEMS COMPANY, MITSUBISHI FUSO TRUCK AND BUS CORPORATION, MITSUBISHI JIDOSHA ENGINEERING, MITSUBISHI HEAVY INDUSTRIES ENGINEERING, MITSUBISHI HEAVY INDUSTRIES AUTOMOTIVE THERMAL SYSTEMS COMPANY	Japan	533
Mazda	MAZDA MOTOR CORPORATION	Japan	718
Suzuki	SUZUKI MOTOR CORPORATION	Japan	371
Subaru	SUBARU	Japan	264

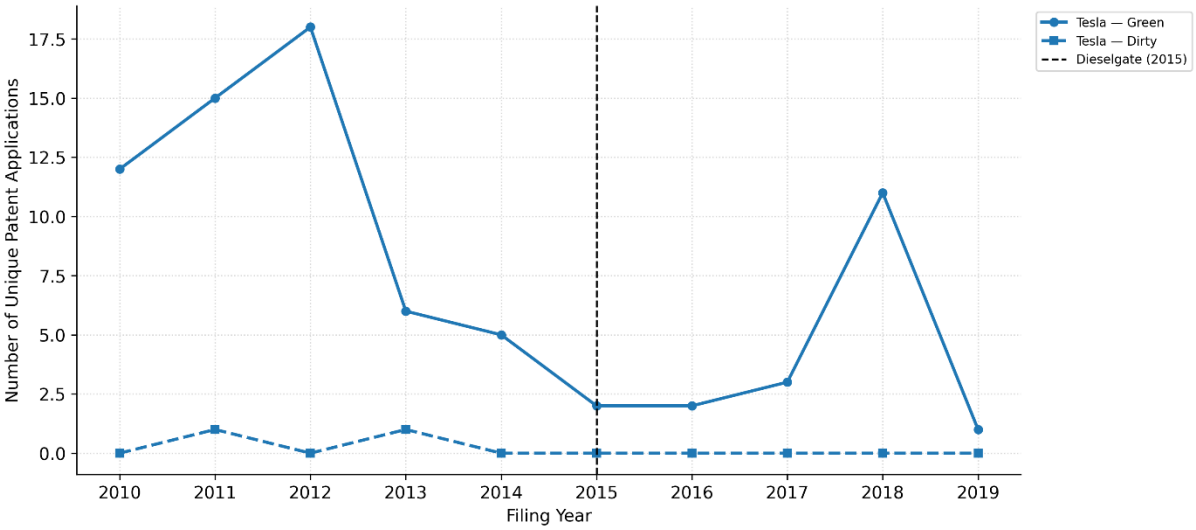
Brand group	**Consolidated psn_name(s) (applicants)**	**Country**	**Patent Counts**
Hyundai-Kia	HYUNDAI MOTOR COMPANY, HYUNDAI MOBIS COMPANY, HYUNDAI AUTRON COMPANY, HYUNDAI DYMOS, HYUNDAI KEFICO CORPORATION, HYUNDAI WIA CORPORATION, HYUNDAI TRANSYS, KIA MOTORS CORPORATION, KIA CORPORATION	South Korea	4,708
Volvo Cars	VOLVO CAR CORPORATION, VOLVO TECHNOLOGY CORPORATION	Sweden	141
Volvo Trucks	VOLVO LASTVAGNAR, VOLVO TRUCK CORPORATION, AB VOLVO PENTA	Sweden	346
Jaguar Land Rover	JAGUAR LAND ROVER, JAGUAR CARS	United Kingdom	261
Fiat	FIAT AUTO, CRF (CENTRO RICERCHE FIAT)	Italy	71
Tesla	TESLA, TESLA MOTORS	United States	77
Sample		**USA**	31,557

The patents in the sample were initially separated into two categories: “dirty” patents, associated with internal combustion engine innovations, and “green” patents, aimed at electric propulsion. For the classification, I used information on a variety of CPC classes derived from the European Patent Office and from Table 1 of Aghion et al. (2016). I drew on resources covering internal combustion engine design and modern electric transportation (van Basshuysen & Schäfer, 2016; Bhatt et al., 2018; Emadi, 2015; Ferrari et al., 2022; Gupta, 2025; Hughes & Drury, 2019; Krishnamoorthy & Parakkat Kesava Panikkar, 2024; Un-Noor et al., 2017), together with the Google NotebookLM tool, to isolate patents related specifically to engines and propulsion. The prompt used for the selection appears in Appendix A1. In Appendix A2, all CPC classes are separated into two groups: dirty and green. Multiple CPC classes can be assigned to a single patent, which complicates the unambiguous determination of the patent's type (green or dirty). I implemented the following rule to unambiguously

determine the type: if there are strictly more “dirty” classes than “green” ones, the patent is considered dirty (majority rule). If the number of “green” classes is at least as large as the number of “dirty” ones, then the patent falls into the green group. The trend in the number of dirty patents per year for Tesla (Figure 1) remains close to zero, which I take as informal evidence of the relative efficacy of the patent classification and the majority rule. For the rest of the makers, the trends (yearly number of dirty and green patents) can be found in Appendix A3.

I also impose additional filters on the patents besides the classification. I restrict the sample to patents filed in the USA, a condition intended to ensure the technology is aimed at the American market. I also clean the patent pool of artificial applications created for PATSTAT's internal purposes and of continuation patents. The final sample amounted to 31,557 patents, 2,501 of which were patents issued by German makers. The patents were aggregated into cells defined by brand group (18), patent type (green or dirty), and year (10). In total, this yielded 360 observations for the patent counts.

Figure 1. Tesla, patent counts.



I also extracted data on forward citations, backward citations, family size, and non-patent literature for each patent. Forward citations are collected as distinct subsequent applications that cite any publication of the focal patent (a single patent application filed in a jurisdiction typically has several publications: a pre-grant publication and a grant publication). I also do not use any fixed citation window, such as the conventional four- or five-year truncation. The standard motivation for truncation is that more recent cohorts have had less time to accumulate citations and therefore appear systematically less cited. However, in my research, the available PATSTAT data were last updated in October 2021, which would have

made the most recent applicable year 2016 – the first year after the scandal broke. This truncation would have complicated the identification of the scandal's long-term impact on the innovation activity of German makers. To address the truncation issue, I use year fixed effects. Backward citations belonging to an individual focal application are defined as distinct applications cited by any publication of the focal application. The deduplication unit (the level at which duplicates collapse into one) is the cited application, not the cited publication. For instance, if a patent application cites both publications of the same prior filing, this counts as one backward citation. Family size is defined as the number of distinct application authorities in which the same invention is filed. Here, I count at the "office" level, not the application level – multiple US filings within the same family contribute only one count toward the family size measure of an individual patent. Non-patent literature references are defined as the number of distinct non-patent references cited by any of its publications.

In total, I use four quality indicators for each patent, which I quantify by summing the number of occurrences of each. Table 2 below presents the descriptive statistics for these indicators. The results show that forward citations, backward citations, and NPL references are strongly right-skewed (see Appendix 4), while the empirical distribution of family size is close to symmetric around its mean, which is almost identical to the median.

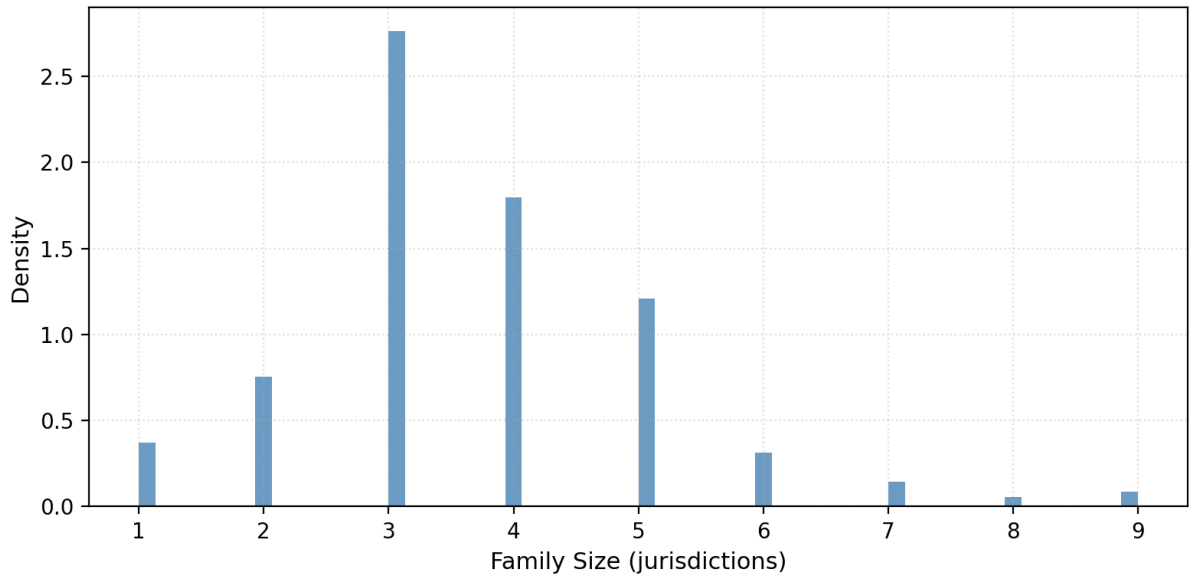
Table 2. Descriptive statistics for the quality indicators.

	N	Mean	Median	SD	Min	Max	Q1	Q3	Share_zeros
Famsize	31,557	3.68	3.0	1.47	1.0	16	3.0	4.0	0.0
Bcit	31,557	13.68	11.0	15.84	0.0	681	5.0	17.0	0.11
Fcit	31,557	4.83	2.0	9.42	0.0	408	0.0	6.0	0.28
Refnpat	31,557	1.09	0.0	3.74	0.0	99	0.0	1.0	0.66

Note: Famsize – family size; Bcit – backward citations; Fcit – forward citations; Refnpat – references to non-patent literature; N – number of observations; SD – standard deviation; Q1/Q3 – first/third quartile; Share_zeros – share of zero-valued observations.

I find this observation consistent with the logic of the family size indicator. First, the number of jurisdictions is strictly restricted by the number of officially recognised states. Second, for firms operating in the same technological field and market, I expect to observe a common set of jurisdictions where patenting activity is essential for all competitors, leading family sizes to concentrate around a certain integer value (4 in this case). Figure 2 supports this claim – the empirical distribution is close to normal.

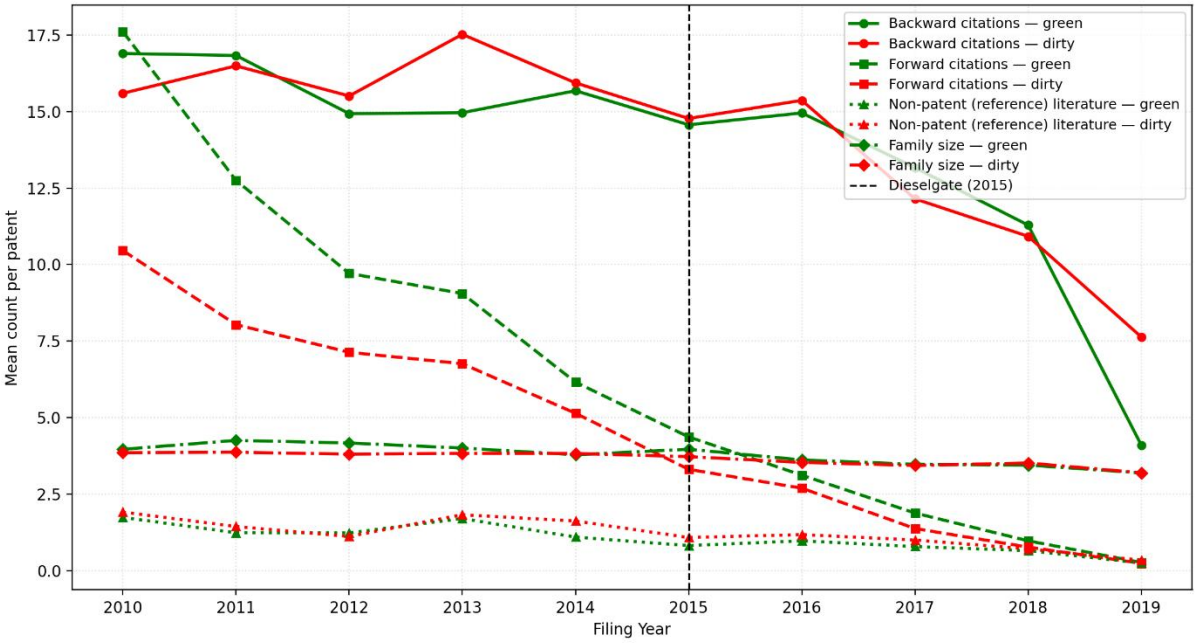
Figure 2. Family size, density plot.



The share of zero values for the references is 66%, which I interpret as evidence that references to non-patent literature are not a feature present in every patent by default. However, every third patent has at least one reference, meaning this property cannot be dismissed as a rare feature confined to a handful of patents. Accordingly, I consider references a systematic indicator of a patent's technical elaboration. Regarding backward citations, I expect to see non-zero values for the majority of patents (89% in this sample), since most innovations are assumed to be based on earlier innovations. The share of zeros for backward citations is also lower than that for forward citations, which I also find realistic. Because the ability to accumulate forward citations depends on how much time has passed since filing, forward citations do not yet exist for 28% of the patents in the sample.

The truncation is visible in the strictly decreasing linear trends shown in Figure 3. Backward citations are also truncated, although they are not subject to the accumulation effect. One possible explanation is the gradual expansion of the PATSTAT database and the addition of citations in subsequent publications.

Figure 3. Each quality indicator's trends classified by the types of technology.



4. Methodology

4.1. The Triple-Difference Estimator

The study is structured around three pairs of comparisons: German makers and non-German makers, dirty and green patents, and the pre-scandal period and the post-scandal period. Because the focus is on the relative difference in how patenting for ICE cars changed compared with patenting for EVs specifically among German makers, net of US-wide trends, this thesis employs the triple difference-in-differences (DDD) framework. Here, 'the USA' refers to the geographic scope of patent applications. Non-German makers therefore form the control group that absorbs trends not directly related to Dieselgate. For the sake of simplicity and interpretability, and following the findings from the papers discussed in the preceding section (Literature review), the impact of the scandal is analysed through aggregated patent counts. The unit of observation is defined as a brand group i in calendar year t filing patents of technology type k, where $k \in$ {green, dirty}. This results in a balanced panel of 18 brand groups × 10 calendar years (2010–2019) × 2 technology types, for a total of 360 observations.

Four quality indicators (family size, forward citations, backward citations, and non-patent literature references) are aggregated by the same principle. One can think of these

measures as proxies for the accumulated quality of patents. For instance, a statistically significant result with respect to forward citations constitutes evidence of a quality shift along the citation axis only, not of an improvement or deterioration in overall patent quality. This analysis cannot judge a patent's overall quality, only its quality along a given indicator. Since no single indicator is exhaustive, all five dependent variables are estimated across identical right-hand-side specifications, and the consistency of results across measures is treated as additional evidence of the effect.

Expressing the DDD in terms of conditional means makes the logic of the triple-difference transparent:

$$\Delta_{\mathrm{DDD}} = \{[E(Y \mid \mathrm{Ger}, \mathrm{D}, \mathrm{Post}) - E(Y \mid \mathrm{Ger}, \mathrm{D}, \mathrm{Pre})] - [E(Y \mid \mathrm{Ger}, \mathrm{Gr}, \mathrm{Post}) - E(Y \mid \mathrm{Ger}, \mathrm{Gr}, \mathrm{Pre})]\}$$

$$- \{[E(Y \mid \mathrm{NGer}, \mathrm{D}, \mathrm{Post}) - E(Y \mid \mathrm{NGer}, \mathrm{D}, \mathrm{Pre})] - [E(Y \mid \mathrm{NGer}, \mathrm{Gr}, \mathrm{Post}) - E(Y \mid \mathrm{NGer}, \mathrm{Gr}, \mathrm{Pre})]\}$$

Here *Ger* = German brand groups; *NGer* = Non-German*; D* = Dirty (ICE patents)*; Gr* = Green (EV/Hybrid patents); *Post* = post-scandal period *($t \geq 2015$); Pre* = pre-scandal period *($t < 2015$).* The outer bracket isolates the change over time in the dirty-versus-green patenting gap within each manufacturer group; the difference between these two group-based differences then removes the component of that change that is common to non-German makers and thus attributable to US-wide trends rather than to the scandal. What remains — Δ_{DDD} — is the German-specific post-scandal shift in dirty patenting relative to green patenting, net of all trends shared with the control group.

4.2. Lin-Linear OLS and Poisson Pseudo-Maximum Likelihood specifications

In this framework, I use two specifications: Poisson Pseudo-Maximum Likelihood (PPML) and the Lin-Linear OLS (Ordinary Least Squares) model. I use PPML as the main specification. This decision is motivated by the distributional properties of the outcome variables documented in the Data section. The linear model serves as the robustness check. However, I start with the Lin-Linear model, since its functional structure provides a more intuitive bridge from the conceptual discussion of the triple difference to its statistical representation:

$$y_{itk} = \alpha_{ik} + \gamma_{tk} + \sum_{t \neq 2014} [\beta_{1t}(\text{German} \times D_t) + \delta_t(\text{German} \times \text{Dirty} \times D_t)] + \varepsilon_{itk}$$

where y_{itk} denotes the outcome variable (patent count or quality indicator) for brand group *i* in year *t* and technology type *k*; α_{ik} are brand-by-type fixed effects that absorb all time-invariant brand–technology heterogeneity, including permanent gaps in firm size, R&D intensity, product portfolio composition, and the baseline green versus dirty patenting split; γ_{tk} are year-by-type fixed effects that absorb aggregate annual shocks common to a given technology type — for instance, economy-wide shifts in battery-technology investment or changes in the regulatory treatment of ICE vehicles. *German* is a binary indicator equal to one for German makers and to zero for non-German makers; *Dirty* is an indicator equal to 1 for ICE-related patents and 0 for EV-related patents, regardless of year or maker group; D_t is a set of year dummies independent of the technological type; β_{1t} captures the year-specific level difference between German and non-German makers common to both technology types; δ_t is the coefficient of interest — the triple-difference — measuring the German-specific dirty versus green differential in year *t* relative to 2014; and ε_{itk} is the error term. Standard errors are clustered at the brand group level, with 18 clusters. Under the Lin-Linear specification, the coefficient δ is interpreted as the average change in the level of the outcome variable attributable to the triple-interaction: the additional unit change in patent counts (or the respective quality indicator) for German makers' dirty patents relative to green patents in the post-scandal period, over and above the analogous change for non-German makers.

In this specification, I use the so-called dynamic coefficients of interest: coefficient δ_t estimates the scandal effect for each year in the sample relative to the reference year 2014 and allows the trajectory of the impact to be traced year by year. The static specification, presented in Section 4.3, collapses the post-scandal period into a single coefficient, thereby providing a summary measure of the average post-scandal impact. The two approaches are complementary: the dynamic event study reveals whether the effect emerged gradually, abruptly, or without any trend, and shows whether it grew over time or faded. The static estimate characterises its average magnitude over the post-scandal window.

A practical advantage of the Lin-Linear model lies in the direct interpretability of the coefficients δ_t as level changes in the outcome variable, making the magnitude of the estimates straightforwardly comparable across years. The main limitation of this approach is that OLS assumes the conditional mean $E(y_{itk} \mid X)$ to be linear in the parameters. This assumption becomes increasingly tenuous when the outcome variable is right-skewed. An important limitation of

this specification concerns heterogeneity. Although the two sets of fixed effects control for the sources of heterogeneity across brand-by-type and year-by-type, one caveat concerns heterogeneity within the treated group: the coefficient δ_t captures an average treatment effect across the three German brand groups — BMW, Daimler, and Volkswagen Group — without distinguishing between them. To the extent that the scandal's direct legal and reputational consequences fell disproportionately on different makers, the estimated δ may understate the effect for one firm and overstate it for another.

The use of PPML with brand fixed effects and clustered standard errors follows the canonical treatment of overdispersed, firm-heterogeneous patent-count panels in Hausman, Hall, and Griliches (1984). It is estimated by pseudo-Poisson maximum likelihood, with standard errors clustered by brand. The equation is given below:

$$E[y_{itk} \mid X] = \exp\Big\{\alpha_{ik} + \gamma_{tk} + \sum_{t \neq 2014} [\beta_{1t}(\text{German} \times D_t) + \delta_t(\text{German} \times \text{Dirty} \times D_t)]\Big\}$$

The main reason behind the choice of PPML as the preferred specification is that the data contain a non-negligible share of zero observations (Table 2). This applies to all cells in the panel. This property of the data makes a log-linear regression of $\ln(1 + C)$ a seemingly natural starting point. That transformation, however, yields inconsistent estimates: by Jensen's inequality, $E[\ln(1 + y) \mid X] \neq \ln(1 + E[y \mid X])$, and the size of this gap depends on X, which under heteroskedasticity systematically biases the coefficients. Santos Silva and Tenreyro (2006) show that PPML resolves this: its estimates are consistent under a single condition — correct specification of the conditional mean as exponential — irrespective of the true distribution of C or the structure of the heteroskedasticity. PPML also handles zero observations directly, without requiring the "+1" shift. The model thus serves a dual purpose: estimating the year-by-year percentage shift in German patenting relative to the control group, and doing so on a zero-laden outcome without losing consistency. Under the PPML specification, δ_t enters the linear index inside the exponential function. The quantity $\exp(\delta_t) - 1$ therefore gives the approximate percentage change in the conditional mean of y_{itk} attributable to the triple-difference in year t relative to the 2014 baseline — a semi-elasticity interpretation that is invariant to the scale of the outcome variable.

4.3. Static Specification and the pre-trend testing

The static specifications replace the vector of year-by-year interaction terms with a single binary indicator $Post_t$, equal to one for all years from 2015 onward, yielding a single coefficient δ that retains its level interpretation: it measures the average post-scandal change in the outcome variable for German makers' dirty patents relative to green patents, net of the analogous change for non-German makers. All fixed effects, standard error clustering, and variable definitions remain unchanged from Section 4.2.

$$y_{itk} = \alpha_{ik} + \gamma_{tk} + \beta_1(\text{German} \times \text{Post}_t) + \delta(\text{German} \times \text{Dirty} \times \text{Post}_t) + \varepsilon_{itk}$$

The structure of the PPML model and the fixed effects are identical to Section 4.2, with the conditional mean specified as exponential. The coefficient δ carries the same semi-elasticity

$$E[y_{itk} \mid X] = \exp\{\alpha_{ik} + \gamma_{tk} + \beta_1(\text{German} \times \text{Post}_t) + \delta(\text{German} \times \text{Dirty} \times \text{Post}_t)\}$$

interpretation as in the dynamic PPML: $\exp(\delta) - 1$ approximates the average percentage change in the conditional mean of y_{itk} over the post-scandal period, relative to the pre-scandal baseline.

A necessary condition for the causal interpretation of δ is that, in the absence of the scandal, the gap between the dirty and green patents would have evolved in parallel (no pre-trend) for German and non-German makers. Since this counterfactual cannot be observed directly, it is assessed indirectly through the pre-scandal period. For each of the ten models, a Wald F-test is performed on the pre-scandal triple-difference coefficients, with the null hypothesis:

$$H_0: \delta_{2010} = \delta_{2011} = \delta_{2012} = \delta_{2013} = 0$$

Under H_0, the German-specific dirty-versus-green differential equals the corresponding control-group differential in every pre-scandal year, yielding DDD coefficients of zero — consistent with the no-pre-trends assumption. All ten regressions (5 outcome variables × 2 specifications) are reported regardless of the pre-trend test outcome. Specifications for which the null hypothesis cannot be rejected at conventional significance levels (10%, 5% and 1%) and for which δ is statistically significant are treated as the preferred results. Specifications that reject the null are retained for completeness but interpreted with caution, as the identification assumption cannot be maintained for those models.

5. Results

I first focus on the dynamic specifications. In Tables 3 and 4, only the DDD coefficients are presented; brand-type and year-type fixed effects are included in the specification but not reported separately. In the first row (Table 4), the PPML model of patent counts supports the hypothesis of a prolonged shifting effect of the scandal and reveals a persistent decrease in the difference between the dirty and green patent counts for German makers. The smallest absolute effect in 2017 is 38%. This result characterizes German makers as systematically having a lower gap between the number of dirty and green patents in favor of green patents in years after the scandal, compared to non-German makers. In 2019, the effect of the scandal is considered zero at the 10% significance level, meaning the difference between the dirty and green patent counts for German makers returns to the level of the pre-scandal year 2014. The statement that, at some point in time, the trajectory returns to the path preconditioned by the technological branch with the largest accumulated innovation stock was proposed in Hémous & Olsen (2021), building on DTC theory from Acemoglu (2002). I cannot assert that 2019 is the year when the effect starts fading and that statistically significant DDD coefficients would never be observed in subsequent years; however, 2019 is the only year in which the dirty-versus-green gap for German makers, compared to 2014, is statistically zero for all models and outcome variables. Appendix 5 presents the graphs for the dynamic DDD-coefficients in a line plot where the horizontal axis is a timeline, and the vertical axis denotes the coefficient values (10 graphs, years from 2010 to 2019). The second model, which measures the influence of family size assuming its non-linear structure, yields results that are quantitatively very similar to those of the model in the first row (when compared year by year). The two models diverge in 2017, when family size loses statistical significance. Another notable result comes from the linear model of patent counts, which reveals a persistent 3-year decrease in the gap between the dirty patent counts relative to green patent counts after the year 2014. This result cannot be considered more persuasive than the PPML model on patent counts, neither in terms of the significance of the coefficients, nor in the time span of the persistent decrease. The value of this model's findings is that the results for the years 2015, 2016 and 2017 are aligned with the results for the PPML model on patent counts for the first three scandal years: the effect took place, and led to the reduction of the gap in favor of green technologies. The linear model of family size suggests the persistence of the technological shift over 4 years after the scandal and, in terms of the duration of persistence, is as strong as the PPML model on patent counts. Although this model (Lin-lin, family size) reveals that over 4 years the difference between the accumulated

numbers of jurisdictions for dirty and green patents separately was lower than in the year 2014 by at least 104 jurisdictions (year 2015), and the inference can be phrased as "After the scandal, German makers increased the intensity of patenting green technologies in international markets in comparison to dirty technologies", I want to note a caveat. Figure 4 shows how the mean number of jurisdictions per patent varies over the years. The means for German makers vary around 4.5 jurisdictions, with yearly standard deviations not exceeding 1.62 (2010). Non-German makers file patents, on average, in 3.5 jurisdictions, with the largest standard deviation of 1.87 (2010). Taking into account the stability of the mean values — visible also in Figure 3 — and the family-size-to-count ratio for the coefficients of the models in rows 6–7 (approximately 3 to 4 for the years 2015–2018), I conclude that much of the change in the accumulated family size gap can be attributed to variability in patent counts, rather than to variability in the quality indicator "family size". Similar attribution can be observed when comparing rows 1 and 2.

Table 3. Dynamic DDD coefficients, before the scandal (2010–2014).

Spec.	Dep. var	2010	2011	2012	2013	2014	F-test p	N	R^2	Maker-tech FE	Time-tech FE
PPML	counts	0.24 *(0.20)*	0.04 *(0.26)*	0.03 *(0.22)*	0.08 *(0.24)*	*0 (ref)*	**0.5334**	360	0.892	Yes	Yes
PPML	famsize	0.38 *(0.27)*	0.12 *(0.25)*	0.12 *(0.20)*	0.06 *(0.21)*	*0 (ref)*	**0.6735**	360	0.899	Yes	Yes
PPML	fcit	0.94*** *(0.34)*	0.38 *(0.28)*	0.28 *(0.21)*	0.20 *(0.32)*	*0 (ref)*	**0.0058*** **	360	0.943	Yes	Yes
PPML	bcit	0.64** *(0.27)*	0.28** *(0.14)*	0.35** *(0.17)*	0.32 *(0.32)*	*0 (ref)*	**0.0023*** **	360	0.915	Yes	Yes
PPML	nptlit	0.90 *(0.64)*	0.65 *(0.68)*	0.95** *(0.43)*	1.11 *(0.72)*	*0 (ref)*	**0.0028*** **	360	0.814	Yes	Yes
Lin-lin	counts	-20.5 *(19.73)*	-7.13 *(14.36)*	-9.40 *(16.20)*	-6.40 *(19.00)*	*0 (ref)*	**0.6751**	360	0.882	Yes	Yes
Lin-lin	famsize	-50.4 *(83.95)*	6.67 *(50.96)*	0.60 *(51.54)*	-10.9 *(64.37)*	*0 (ref)*	**0.8947**	360	0.882	Yes	Yes
Lin-lin	fcit	28.3 *(165.57)*	152.7* *(83.71)*	18.3 *(153.08)*	-3.53 *(142.61)*	*0 (ref)*	**0.2057**	360	0.707	Yes	Yes
Lin-lin	bcit	-24.3 *(340.75)*	48.9 *(199.64)*	44.4 *(280.17)*	-84.1 *(345.26)*	*0 (ref)*	**0.9847**	360	0.855	Yes	Yes
Lin-lin	nptlit	7.93 *(86.94)*	45.9 *(94.43)*	83.0 *(56.89)*	95.1 *(83.42)*	*0 (ref)*	**0.1398**	360	0.640	Yes	Yes

Table 4. Dynamic DDD coefficients, after the scandal (2015–2019).

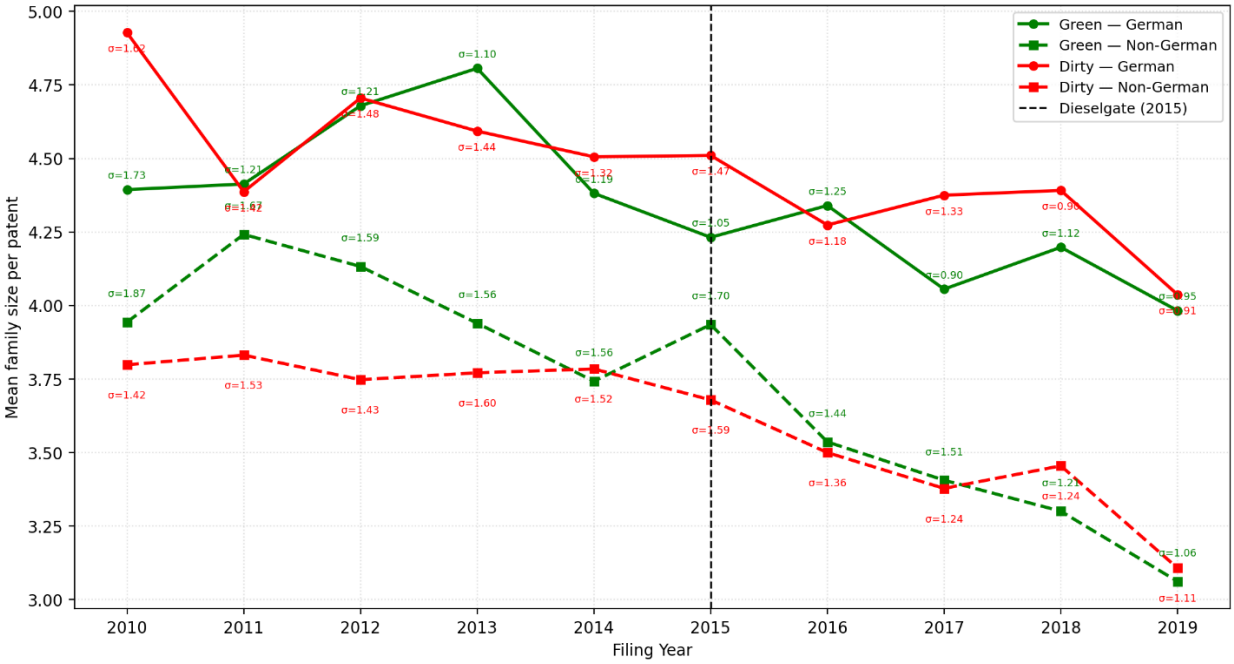
Spec.	Dep. var	2015	2016	2017	2018	2019	N	R^2	Maker-tech FE	Time-tech FE
PPML	counts	-0.55*** *(0.13)*	-0.71*** *(0.26)*	-0.48* *(0.25)*	-0.87*** *(0.30)*	-0.29 *(0.36)*	360	0.892	Yes	Yes

Spec.	Dep. var	2015	2016	2017	2018	2019	N	R^2	Maker-tech FE	Time-tech FE
PPML	famsize	-0.44*** *(0.12)*	-0.73*** *(0.25)*	-0.41 *(0.27)*	-0.89*** *(0.29)*	-0.31 *(0.41)*	360	0.899	Yes	Yes
PPML	fcit	-0.40* *(0.21)*	-1.08*** *(0.29)*	-0.49 *(0.30)*	-0.61* *(0.33)*	-0.32 *(0.49)*	360	0.943	Yes	Yes
PPML	bcit	-0.19** *(0.08)*	-0.15 *(0.18)*	0.07 *(0.28)*	-0.55* *(0.29)*	0.04 *(0.35)*	360	0.915	Yes	Yes
PPML	nptlit	-0.33 *(0.46)*	-0.50 *(0.56)*	-0.03 *(0.52)*	-0.39 *(0.57)*	0.81 *(0.52)*	360	0.814	Yes	Yes
Lin-lin	counts	-34.9** *(12.94)*	-52.4** *(18.48)*	-35.7** *(14.45)*	-58.2 *(37.88)*	-9.27 *(23.08)*	360	0.882	Yes	Yes
Lin-lin	famsize	-104.2*** *(32.98)*	-194.1*** *(61.03)*	-113.0** *(40.20)*	-250.0* *(142.88)*	-67.2 *(79.12)*	360	0.882	Yes	Yes
Lin-lin	fcit	-75.2 *(83.41)*	-158.4 *(105.52)*	-24.8 *(112.61)*	-9.53 *(132.83)*	11.3 *(145.57)*	360	0.707	Yes	Yes
Lin-lin	bcit	-314.8** *(137.58)*	-468.6* *(246.77)*	-31.8 *(246.92)*	-365.9 *(381.30)*	144.3 *(361.84)*	360	0.855	Yes	Yes
Lin-lin	nptlit	-19.4 *(57.66)*	-57.4 *(55.74)*	-1.40 *(54.88)*	-28.6 *(71.57)*	84.9 *(70.73)*	360	0.640	Yes	Yes

*Note: only the dynamic DDD coefficients δ_t are reported (the German×year terms β_{1t} are omitted). The F-test p column (before-the-scandal part) reports the joint pre-trend F-test on the pre-period δ_t, with p-value. Significance: *** p<0.01, ** p<0.05, * p<0.10.*

To briefly summarize the results for the dynamic models, the specifications on patent counts and family size show the highest degree of scandal-effect persistence for both PPML and linear models: statistically significant impacts lasting 3 to 4 years beginning in 2015. For these four models (PPML on counts, PPML on family size, and their linear counterparts), the effects in each year are negative, meaning the gap between dirty and green patent counts, and the related quality gap, was reduced after the scandal in favor of green technologies. This effect is consistent with the predictions of DTC theory. All models show an insignificant effect in the year 2019, meaning the gap returned to the level of 2014. Forward and backward citations do not reveal the effect for longer than two years immediately following the scandal. According to both model specifications, patent quality along the non-patent-literature axis did not undergo any structural redistribution in favor of green technologies.

Figure 4. Family size, average over technology type, makers and years.



I next analyze the results of the pre-trend testing conducted through the F-test methodology. The null hypothesis about the absence of a pre-trend is rejected at the 1% level for the PPML models on forward and backward citations and non-patent literature. For the linear model on non-patent literature, the F-test statistic's p-value is close to the 10% level, but exceeds it. For the rest of the models, the probabilities of the first error are far from the thresholds, exceeding 0.5 for the models on patent counts, family size, and backward citations (linear). As a result, for 7 of the 10 models, the pre-assumption of coinciding gaps in the production of dirty and green technologies holds. Hence, the identification of the scandal effect for German makers is feasible. For the sake of consistency, I use the results of the static analysis for all regressions.

Table 5 presents the results of the static triple difference-in-differences analysis. The rows colored in green show the results for the models for which the pre-trend is flat and the scandal impact is significant over the entire time span after the scandal. For the PPML model on patent counts, the gap was reduced after the scandal by an average of 48.8% in favor of green patents, or by 3.5 standard deviations (0.67 / 0.20). The economic meaning of this result is that German makers' relative production of green technologies was incentivized by the scandal to a greater extent than for non-German makers. In other words, the collective reputation spillover, present already in 2015, persisted over the following four years and was reflected in the innovation behaviour of German makers. Here, I must note that I am not in a

position to discuss that spillover over the 2015–2019 period from a commercial or financial perspective. Similar results are obtained for family size modeled non-linearly. The gap between the accumulated number of jurisdictions attributed to dirty and green patents is reduced by 50%. This would suggest that German makers started internationalizing their green technologies more intensively than their competitors did; however, as I previously mentioned, this numerical result is largely attributable to variation in patent counts, since the average family size is stable across the treatment and control groups and across the types of technologies. For the Lin-linear model on family size, the same interpretive logic applies.

Table 5. Static models results.

Spec.	Dep. var	Window	δ (DDD)	p(δ)	pre-trend p	Parallel trends	N	R^2	Maker-tech FE	Time-tech FE
PPML	counts	Post ≥ 2015	**-0.67*** ***(0.20)***	0.0009	0.5334	parallel (no pre-trend)	360	0.890	Yes	Yes
PPML	famsize	Post ≥ 2015	**-0.70*** ***(0.26)***	0.0064	0.6735	parallel (no pre-trend)	360	0.896	Yes	Yes
PPML	fcit	Post ≥ 2015	**-0.99*** ***(0.25)***	0.0001	0.0058	PRE-TREND (rejected)	360	0.942	Yes	Yes
PPML	bcit	Post ≥ 2015	**-0.45*** ***(0.27)***	0.0957	0.0023	PRE-TREND (rejected)	360	0.913	Yes	Yes
PPML	nptlit	Post ≥ 2015	**-0.92*** ***(0.49)***	0.0611	0.0028	PRE-TREND (rejected)	360	0.805	Yes	Yes
Lin-lin	counts	Post ≥ 2015	**-29.43** ***(17.52)***	0.1114	0.6751	parallel (no pre-trend)	360	0.880	Yes	Yes
Lin-lin	famsize	Post ≥ 2015	**-134.91**** ***(57.39)***	0.0311	0.8947	parallel (no pre-trend)	360	0.879	Yes	Yes
Lin-lin	fcit	Post ≥ 2015	**-90.48** ***(149.25)***	0.5524	0.2057	parallel (no pre-trend)	360	0.705	Yes	Yes
Lin-lin	bcit	Post ≥ 2015	**-204.35** ***(230.36)***	0.3874	0.9847	parallel (no pre-trend)	360	0.853	Yes	Yes
Lin-lin	nptlit	Post ≥ 2015	**-50.76** ***(30.29)***	0.1121	0.1398	parallel (no pre-trend)	360	0.635	Yes	Yes

*Note: δ (DDD) is the static triple-difference coefficient; p(δ) is its p-value. "Pre-trend p" reports the joint pre-trend F-test p-value. Significance: *** p<0.01, ** p<0.05, * p<0.10.*

I also emphasize that the DDD coefficient from the Lin-linear model on patent counts is close to significance at the 10% level (0.1114). For the rest of the models, all of which measure the aggregated quality of the filed patents, either the identification strategy is invalidated by pre-trends, or no significant difference from the levels of the pre-scandal period is revealed. I conclude that the aggregated quality of the patents filed was not affected by the scandal. However, this does not mean the average quality of patents is equal across both technology types: a lower number of filed patents can be compensated by higher patent quality, and, conversely, a lower average quality per patent is associated with a larger sample.

6. Conclusion

In this thesis, I analyze the consequences of Dieselgate for the innovation policy of German makers in the USA. The research question can be formulated as "Did the relative intensity of the production of green technologies grow after the scandal and because of the scandal?" The empirical findings contribute to understanding how large producers behave after being publicly exposed for non-compliance with environmental regulations. Policymakers gain insight into the extent to which large (about USD 35 billion) fines and settlements incentivize automakers to actually change their innovation policy. I conduct quantitative analysis and use patent counts as the measure of innovation activity. I also use qualitative indicators of each patent (forward and backward citations, family size, non-patent literature references) to extend the analysis beyond pure patent counts. A single observation is the number of patents or the accumulated value of a certain quality indicator associated with a brand group in a particular year. The PATSTAT database served as the source of data on patents, patent properties, and enterprises. I use non-German makers in the USA as the control group that absorbs trends in the American market and worldwide that impact the entire automotive industry. The 3-dimensional analysis (green and dirty technologies; German and non-German makers; pre-scandal and post-scandal periods) motivates the choice of the triple difference-in-differences model as the identification specification. I also use two different assumptions about the data distribution to answer the research question: non-linearity and linearity. The former is tested through the PPML specification and the latter through the linear regression model.

Looking at the results in aggregate, the picture that emerges is not one of a full abandonment of ICE technologies in favor of green ones, but a redirection that is real, and yet temporary. Under the family size indicator, the persistent reduction of the dirty-green gap points to a period of sustained relative growth in green patenting: German makers filed proportionally more green patents than their non-German competitors did, and this greater output was reflected in a broader cumulated international filing footprint, even though the average scope per individual patent did not change materially. The same conclusion follows from patent counts, where the extensive margin moved decisively toward green technologies for three to four years after 2015, before returning to the pre-scandal baseline by 2019. What did not move is the intensive margin. None of the citation-based outcomes — forward citations, backward citations, or references to non-patent literature — support a comparable causal claim: either the pre-trend test rejects the parallel-trends assumption outright, or the estimated post-scandal effect is statistically indistinguishable from zero and non-persistent. This means the scandal did

not detectably alter how technologically important German makers' green patents were relative to their dirty ones. The quality and knowledge content of that patenting appears to have followed a scandal-independent trajectory. The fading of the volume effect by 2019 is consistent with the path-dependence mechanism: because the dirty sector carries the larger historical knowledge stock, the scandal narrowed the gap for a few years but was not powerful enough to overturn the self-reinforcing advantage of ICE technology. The inventive effort could drift back toward its pre-scandal path. However, here the study reaches its limits, and I cannot assert the result would have been different for the years after 2019.

A related point is made by Knittel (2011), who shows that, absent a binding regulatory or price signal, manufacturers redirect technical effort toward attributes consumers already value — power and weight — rather than toward fuel economy: holding 1980 vehicle size and power constant, fuel economy could have risen by roughly 58% by 2006, yet only about 15% of that potential was realized.

Several extensions could sharpen this analysis further. First, since the scandal concerned defeat devices installed specifically in diesel engines, disaggregating the dirty category into diesel-specific and gasoline-specific ICE patents would allow a direct test of whether the redirection documented above was concentrated in diesel-related technologies, as the scandal narrative would suggest, or spread more broadly across combustion-engine innovation as a whole. Second, the number of claims contained in a patent application — a count variable that delineates the scope of legal protection an applicant is seeking for an invention, and that the literature review already identifies as one of the traditional indicators of patent value alongside family size and citations — was never incorporated into the empirical analysis as a dependent variable. Adding it to the quality battery would test whether the scandal affected the breadth of protection sought for green inventions even in years when the citation counts did not move. Third, the analysis is restricted to patent applications filed with the USPTO (US Patent and Trademark Office); the same triple-difference design could be applied to EPO or German applications, which would speak directly to whether the redirection found for the US market also holds for the European market. This is a domestic market for German makers, and the EU (European Union)-level regulatory reaction to Dieselgate might have produced a different response. Finally, within the green category itself, it would be informative to separate patents specific to fully electric propulsion from those specific to hybrid drivetrains, since the two technologies imply a rather different degree of departure from the internal combustion engine.

Pooling them may mask whether German makers moved toward a genuine electrification path or toward incremental hybridization.

For policymakers, the main takeaway is that a large reputational and financial shock can move automakers' innovation effort toward green technologies. But the effect is neither permanent nor deep: it operates mainly through a reduced gap between dirty and green filings rather than through a lasting change in the underlying quality or science content of green innovation, and it fades within about four years. This suggests that ex-post reputational and legal punishment for non-compliance is not, by itself, a substitute for standing price or regulatory instruments — such as fuel-economy standards, subsidies or a carbon tax — that keep the relative return to green innovation elevated over time. Scandal-driven redirection of the kind documented here appears to be a useful complement to such instruments, rather than a replacement for them.

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Appendix A1

1. Build ICE automobile patterns.

From the provided CPC scheme, select all classes/groups/subgroups that are related to internal combustion engine road vehicles (automobiles) and their supporting systems. Include, for example (non-exhaustive guidance):

Engine itself: reciprocating piston, rotary, multi-fuel, etc.

Air, fuel, ignition, exhaust, lubrication, cooling, starting systems.

Engine mounts, vibration damping, cranks, pistons, turbo/supercharging, EGR, after-treatment, exhaust gas recirculation, silencers.

Driveline parts when they are clearly dedicated to ICE installation.

Constraints:

Only consider applications to automobiles with ICE powertrains (no aircraft, ships, stationary engines, hand tools, lawn mowers, etc.).

It is allowed to include “servicing”, complementary, or auxiliary systems as long as they are clearly part of or dedicated to an ICE road vehicle (e.g., exhaust after-treatment for passenger cars; engine brake control for trucks).

Do not include classes that are generic to any prime mover (e.g., general transmissions, generic pumps, generic bearings) unless the CPC wording or notes clearly restrict them to ICE automobiles.

Using only such ICE automobile related CPC classes, produce a Python list of regular expressions named ice_patterns.

Each element should be a raw string regex that matches the beginning of the CPC symbol.

Use anchors and simple patterns, e.g. r'^F02B' or r'^B60K\s*15/'.

Keep the level of detail consistent: if most entries are at main group level, only go down to subgroup level when needed to exclude EV/HEV specific material.

The output format must be valid Python:

python

```
ice_patterns = [
    r'^F02B',  # Internal-combustion piston engines (automobiles)
    r'^F02D',  # Controlling combustion engines (engine & vehicle-level)
    r'^F01N',  # Exhaust apparatus / silencers for ICE vehicles
    ...add all other relevant ICE-automobile patterns here...
```

]

Do not include any EV/HEV related CPC ranges. Use only the CPC classes from the provided schemes in sources, and take into account the info about engines from the sources.

2. Build EV/HEV automobile patterns.

From the same CPC scheme, select all classes/groups/subgroups that are specifically related to electric vehicles (EV) and hybrid electric vehicles (HEV) used as road vehicles. Include, for example (non-exhaustive guidance):

Electric road vehicle propulsion and control classes such as electrically propelled vehicles, charging, bidirectional power transfer, etc.

Road vehicle specific electric/hybrid arrangements (electrical propulsion units arrangement) and plural diverse prime movers, hybrids.

Other clearly EV/HEV specific vehicle-side systems (on-board charging, high-voltage distribution in vehicles, EV specific auxiliaries) when their wording restricts them to electrically propelled or hybrid road vehicles.

Constraints:

Focus on automobile applications; skip classes explicitly directed only to rail, ships, aircraft, or stationary equipment.

Do not include generic battery or fuel cell construction classes unless they are explicitly limited to vehicle drive trains or EVs in their definition.

If there is nothing in a group that can be reasonably linked to EV/HEV road vehicles, do not include that group at all.

Using only such EV/HEV automobile related CPC classes, produce a Python list of regular expressions named ev_patterns.

Use the same style: anchored raw string regexes on the CPC prefix or main group.

Avoid overlap with any element of ice_patterns; if a class covers both ICE and EV, do not include it in either list (leave it unclassified).

The output format must be valid Python:

python

```
ev_patterns = [
    r'^B60L',    # Electrically-propelled vehicles, charging, control
    r'^B60K\s1/',  # Arrangement of electrical propulsion units (EV)
    r'^B60K\s6/',  # Arrangement of plural diverse prime-movers (HEV)
    ...add all other relevant EV/HEV-automobile patterns here...
```

]

3. Non-overlap and cleaning rules.

Apply these rules strictly when constructing both lists:

No intersection: A CPC pattern must not appear or be implied in both ice_patterns and ev_patterns. If a CPC group explicitly mixes ICE and EV/HEV with no clear separation, leave it out of both lists.

Automobile focus: Exclude patterns that are clearly limited to non-road-vehicle applications (ships, aircraft, stationary).

Remove previous content: Discard any existing values of ice_patterns and ev_patterns from earlier turns and redefine them from scratch according to these instructions.

4. Final output format.

Return only the following Python code block, with no extra commentary or explanation:

python

ice_patterns = [

  # ...final ICE regex patterns...

]

ev_patterns = [

  # ...final EV/HEV regex patterns...

]

Use only the CPC classes from the provided schemes in sources, and take into account the info about engines from the sources.

Appendix A2

Table A2.1. Patent classification, CPC classes.

Dirty	**Green**
F01B	B60L
F01L	B60K 1/
F01M	H01M 2220/20
F01N	H02K 7/006
F01P	H02J 7/00
H01T	H02J 2207/20
H02J 7/14	H02J 50/
F02B	F01P 2050/30
F02D	F03D 9/32
F02F	F03G 7/081
F02G 5/	F16D 2500/1064
F02M	F16H 3/724
F02N	F16H 57/02034
F02P	F16H 57/0476
F04B 17/05	F16H 2200/0021
F04B 35/002	F25B 2240/10
F04B 41/04	F28D 2021/0043
F04C 15/062	B60H 1/00278
F04C 29/122	B60H 1/143
F05B 2203/06	B60K 6/

Dirty	**Green**
F05D 2220/40	B60W 20/
F15B 2211/20523	B60W 10/08
F16C 3/04	B60W 10/24
F16C 3/06	B60W 10/26
F16C 3/14	B60W 10/28
F16C 3/16	F01P 2050/24
F16C 3/20	F16D 2500/1066
F16C 3/22	F16H 3/727
F16C 5/00	F16H 2061/6602
F16C 7/00	
F16C 9/00	
F16F 15/24	
F16F 15/26	
F16F 15/283	
F16H 57/0475	
F16H 59/74	
F16H 61/21	
F16H 2059/74	
F16H 2063/50	
F16H 2500/1062	
F16H 2500/1068	
F16J 1/00	

Dirty	**Green**
F16J 7/00	
F16J 9/00	
F16J 10/00	
F16M 1/021	
F16M 5/00	
F16N 39/08	
F23Q 7/001	
F23Q 2007/	
F25B 2327/001	
F28D 11/025	
F28D 2021/0026	
F28D 2021/0082	
F28D 2021/0087	
F28D 2021/0089	
F28D 2021/0091	
F28D 2021/0094	
B60K 5/	
B60K 13/	
B60K 15/	
B60H 1/025	
B60H 1/18	
B60H 1/22	

Dirty	**Green**
B60Q 1/2688	

Appendix A3

Figure A3.1. German makers, patent counts.

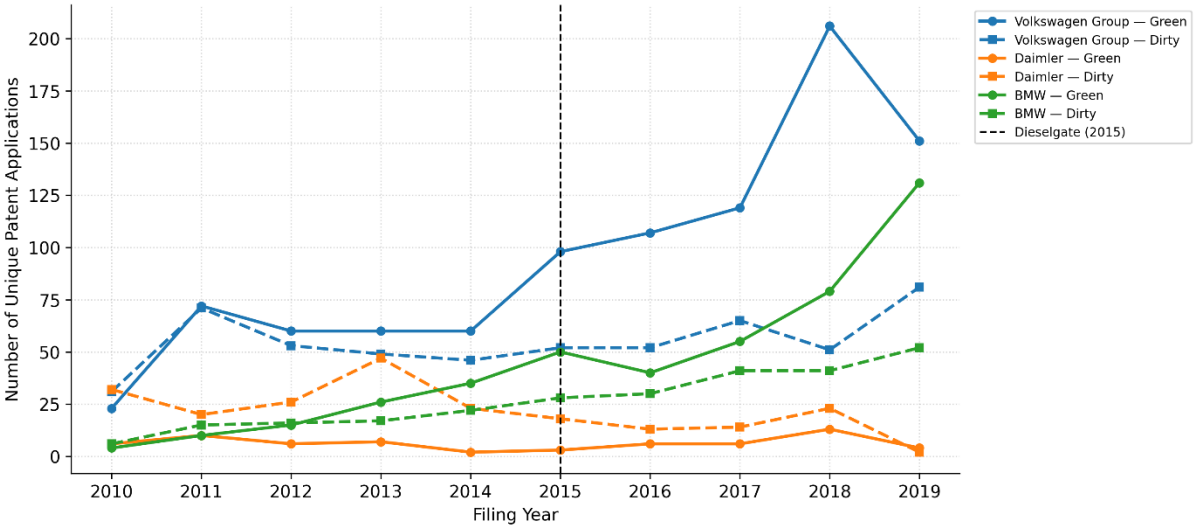


Figure A3.2. American makers, patent counts. Tesla is excluded.

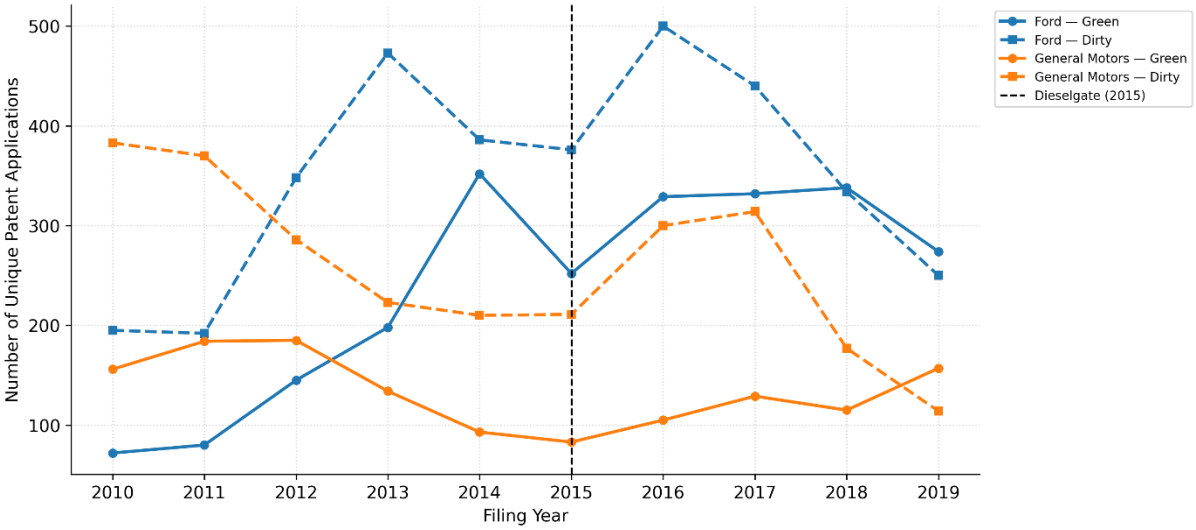


Figure A3.3. Japanese makers, patent counts.

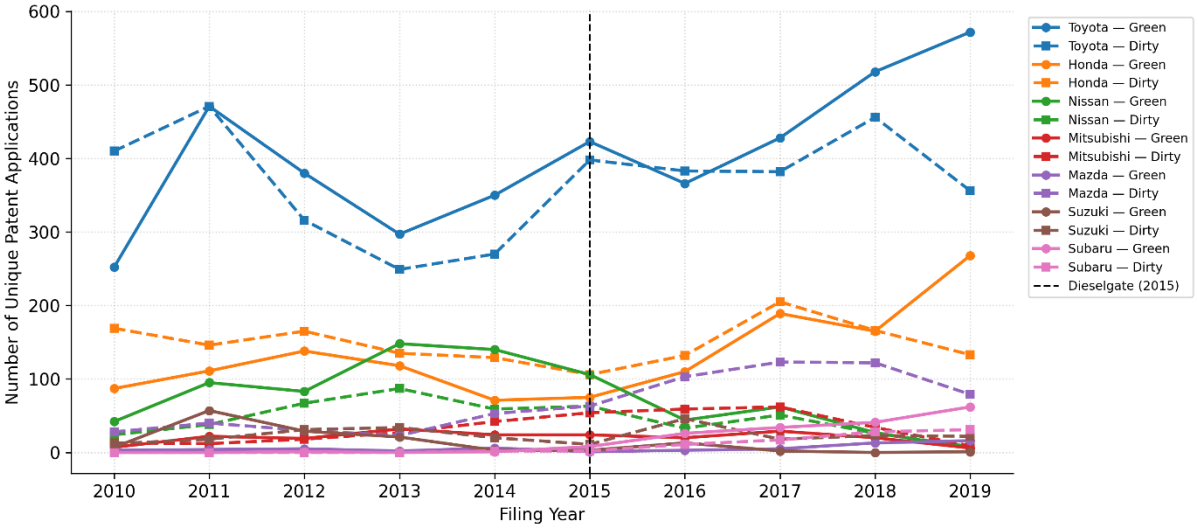


Figure A3.4. Korean makers, patent counts.

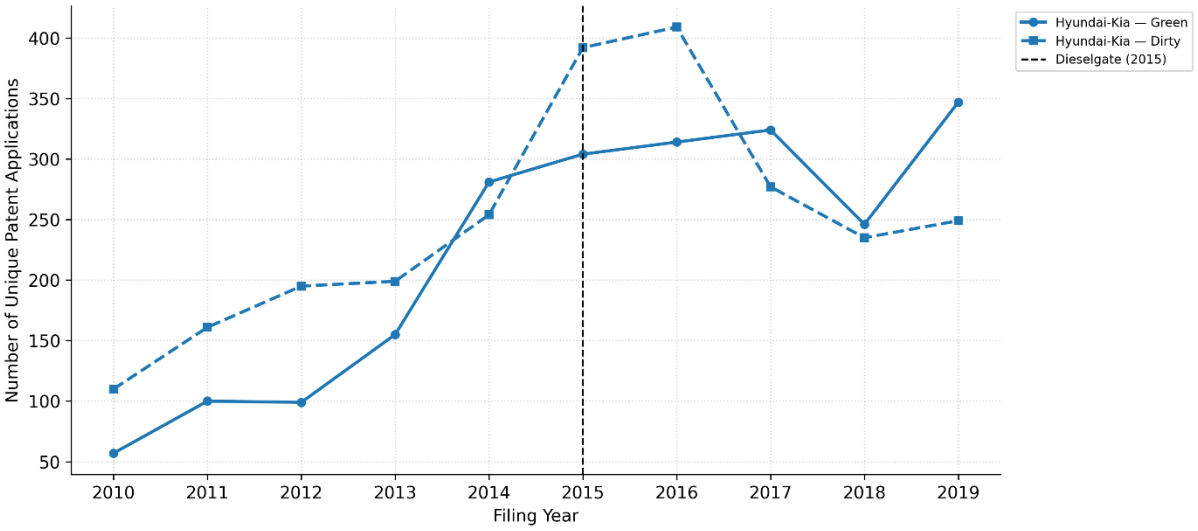


Figure A3.5. Swedish makers, patent counts.

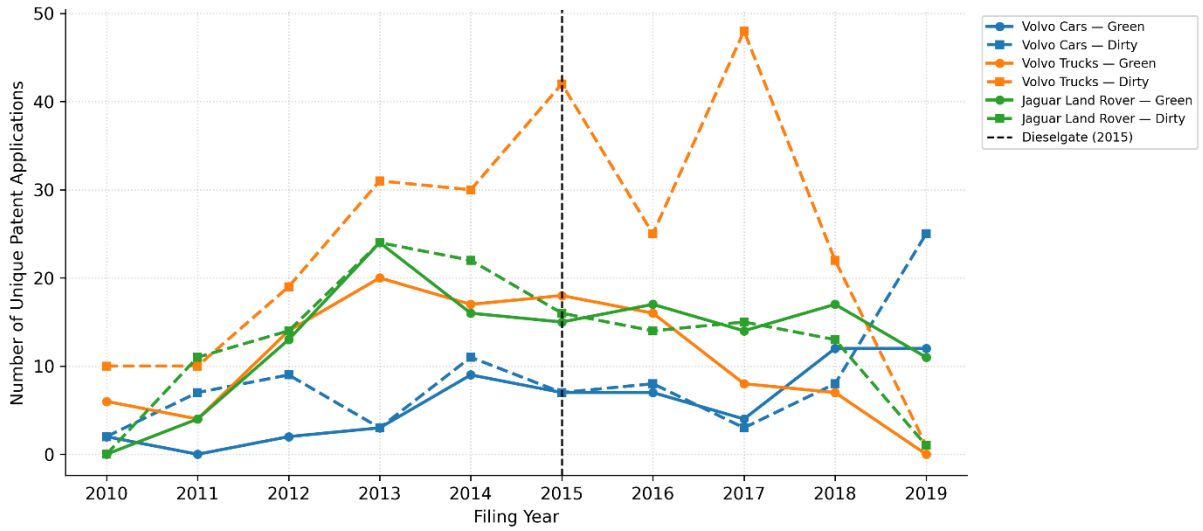
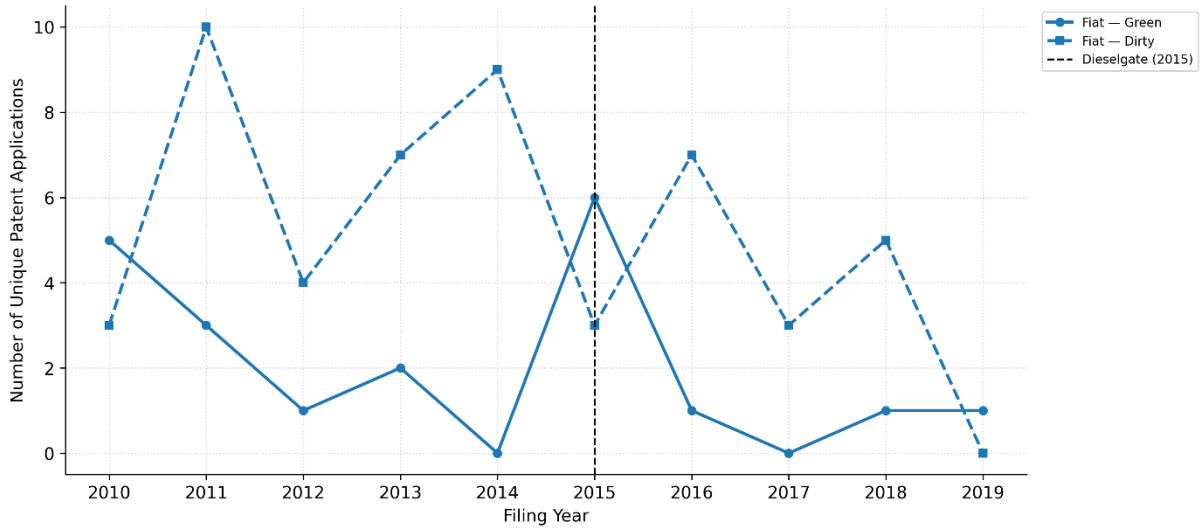
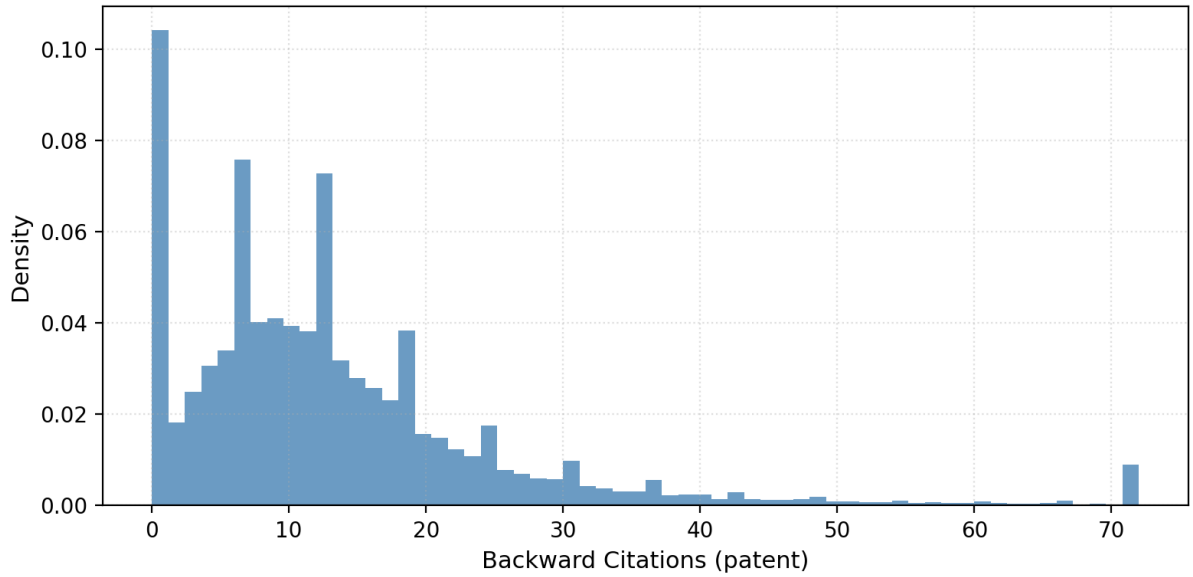


Figure A3.6. Italian makers, patent counts.

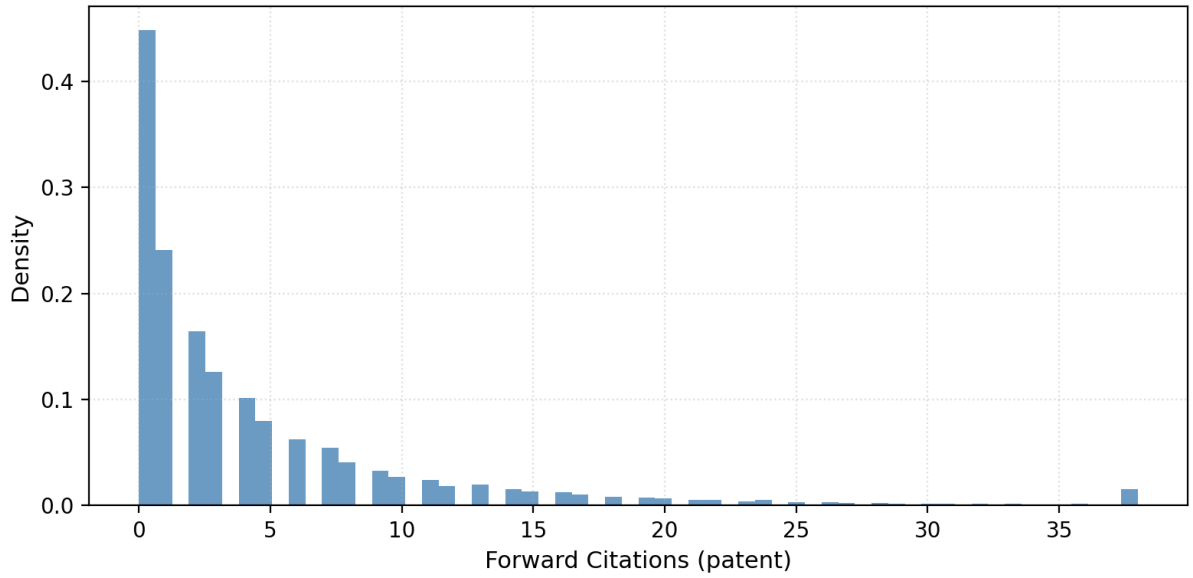


Appendix A4

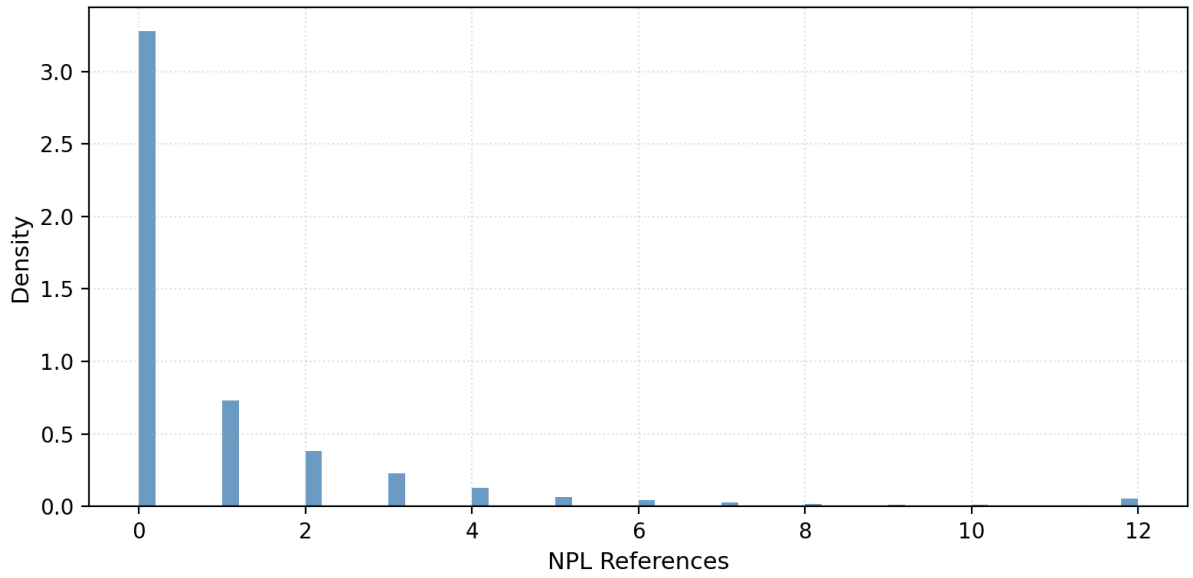
A4.1. Backward citations density.



A4.2. Forward citations density.



A4.3. References to non-patent literature.



Appendix A5

Figure A5.1. PPML Model of the Triple-Difference Effect on Patent Counts.

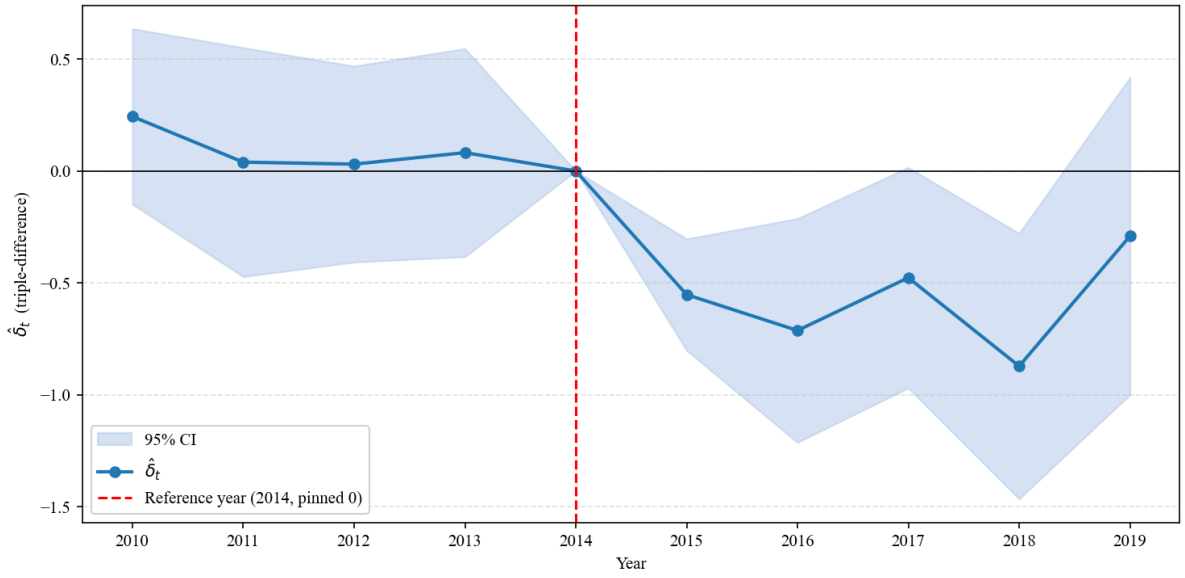


Figure A5.2. Lin-Lin Model of the Triple-Difference Effect on Patent Counts.

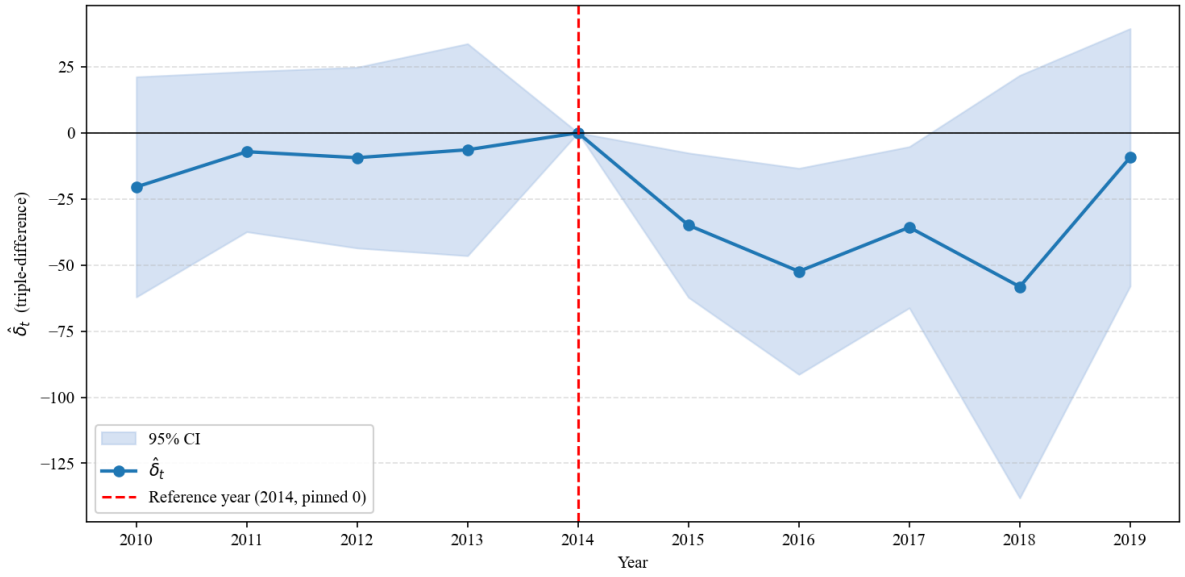


Figure A5.3. PPML Model of the Triple-Difference Effect on Family Size.

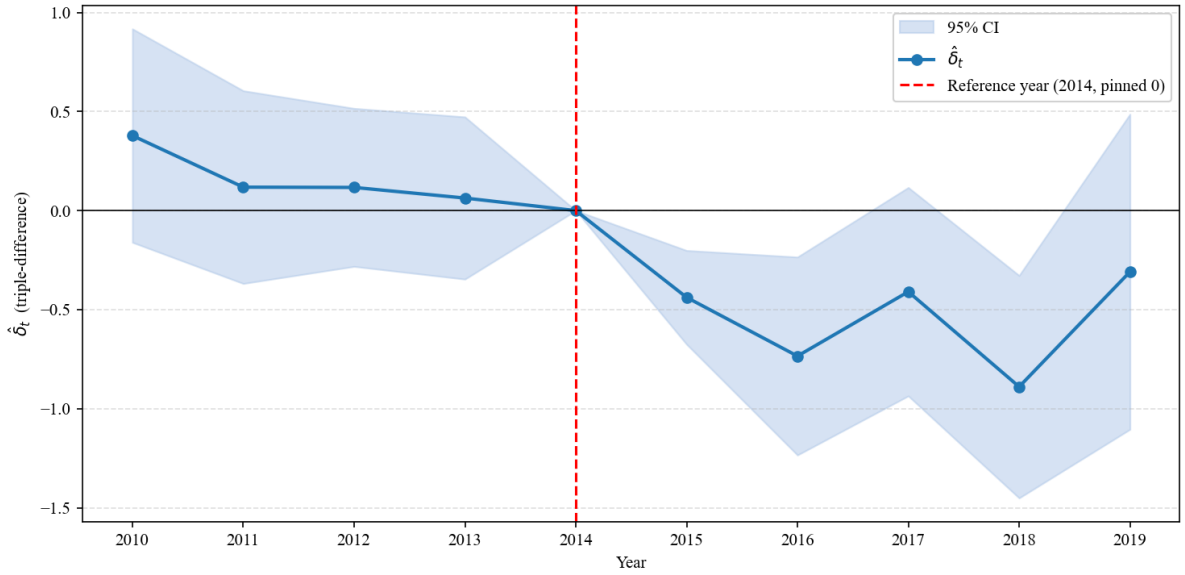


Figure A5.4. Lin-Lin Model of the Triple-Difference Effect on Family Size.

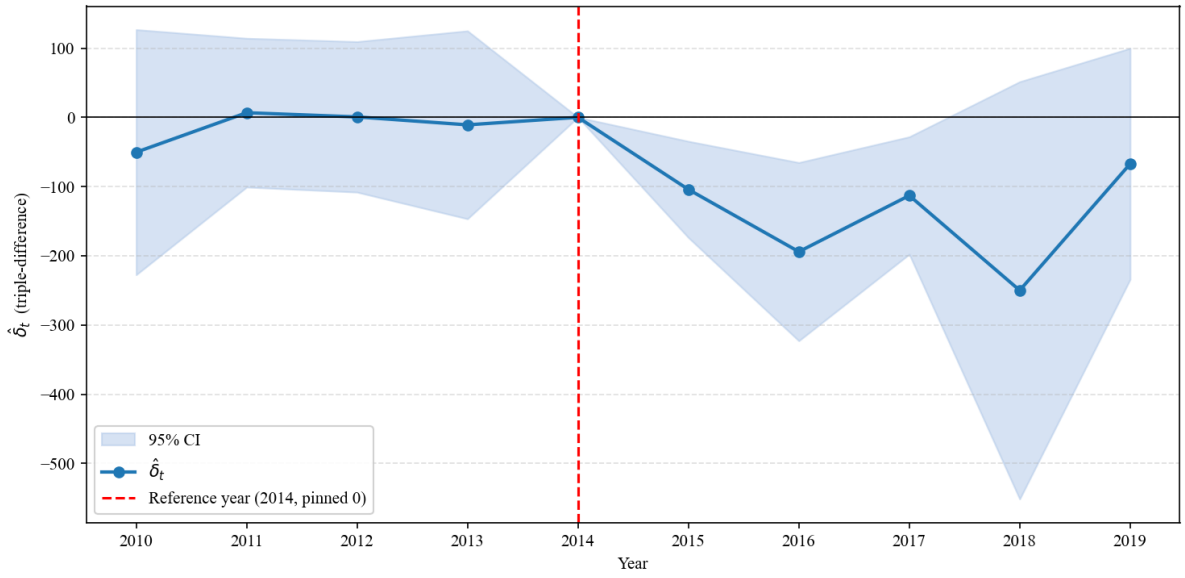


Figure A5.5. PPML Model of the Triple-Difference Effect on Forward Citations.

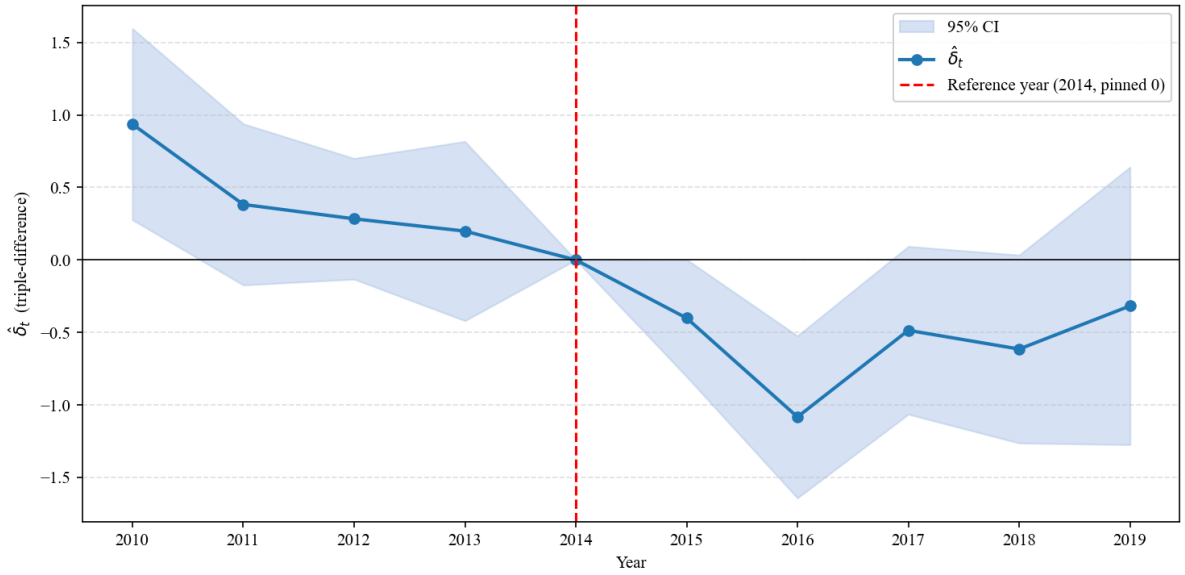


Figure A5.6. Lin-Lin Model of the Triple-Difference Effect on Forward Citations.

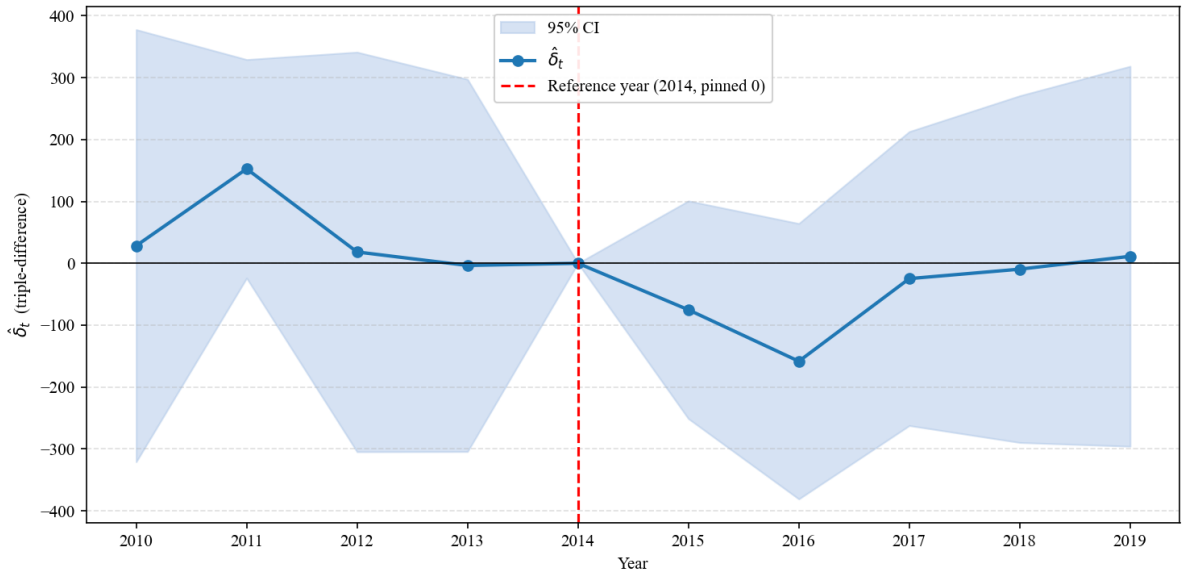


Figure A5.7. PPML Model of the Triple-Difference Effect on Backward Citations.

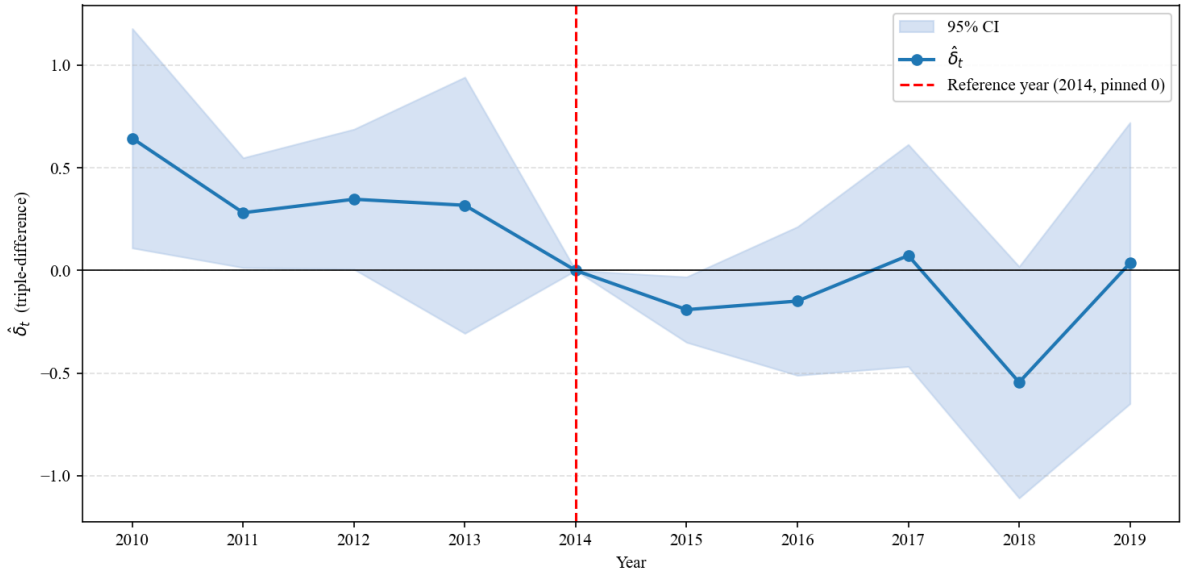


Figure A5.8. Lin-Lin Model of the Triple-Difference Effect on Backward Citations.



Figure A5.9. PPML Model of the Triple-Difference Effect on Non-Patent-Literature Citations.

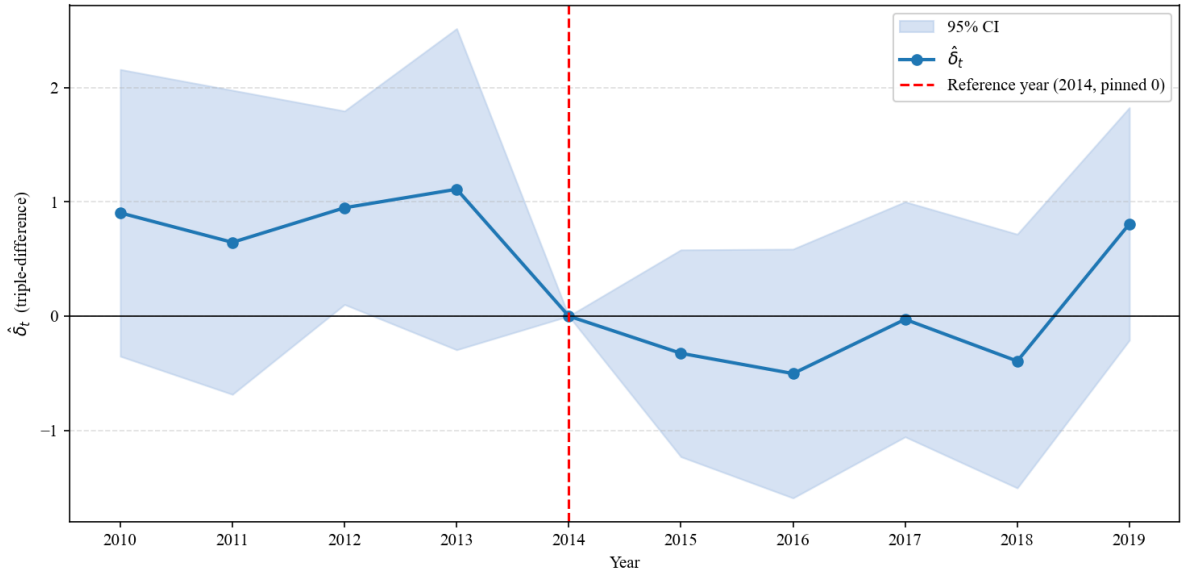
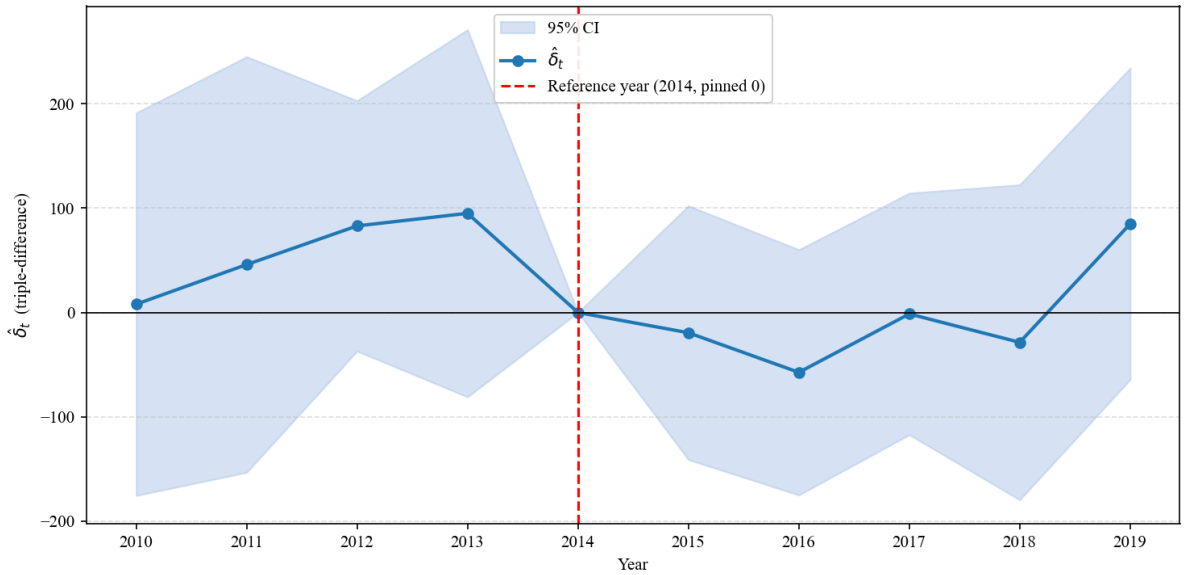


Figure A5.10. Lin-Lin Model of the Triple-Difference Effect on Non-Patent-Literature Citations.



Affidavit

I, Andrey Gusak (Name), student registration number: 23369954, hereby confirm that I completed the submitted work independently and without the unauthorized assistance of third parties and without the use of undisclosed and, in particular, unauthorized aids. This work has not been previously submitted in its current form or in a similar form to any other examination authorities and has not been accepted as part of an examination by any other examination authority.

Where the wording has been taken from other people's work or ideas, this has been properly acknowledged and referenced. This also applies to drawings, sketches, diagrams and sources from the Internet.

In particular, I am aware that the use of artificial intelligence is forbidden unless its use as an aid has been expressly permitted by the examiner. This applies in particular to such programs that could process the task of the examination or parts thereof instead of me, for example chatbots (such as ChatGPT) and other methods suitable for this purpose.

Furthermore, I am aware that working with others or by means of social or other media represents the unauthorized assistance of third parties within the above meaning, if group work is not expressly permitted. Each exchange of information with others during the examination, with the exception of examiners, persons authorized by them and invigilators, about the structure or contents of the examination or any other information such as sources is not permitted. The same applies to attempts to do so. Proofreading of the work by third parties with the aim of correcting linguistic deficiencies is permitted.

Any infringements of the above rules constitute fraud or attempted fraud and shall lead to the examination being graded "fail" ("nicht bestanden"). In the event of multiple or particularly drastic breaches of the rules, further sanctions may be imposed by the Examination Board (Prüfungsausschuss).

Nuremberg, 22.07.2026 

Place, Date Signature